

THE CENTRIOLE AS A UNIDIRECTIONAL RATCHET OF CELLULAR IDENTITY AND AGING

Centriole Reset: Testing Structural Organelle Rejuvenation
in Somatic Cells — Theory, Protocol & Peer-Review Dossier



old centriole → elimination → verified de novo biogenesis → young centriole

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Falsifiable research program · every claim marked [E]/[I]/[H]/[P]

The Centriole as a Unidirectional Ratchet of Cellular Identity and Aging

Centriole Reset: Testing Structural Organelle Rejuvenation in Somatic Cells

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Abstract

The organism may age because the oldest centrioles — carrying the most errors — accumulate in stem cells: division tempo falls and niche-signal perception falls, so tissue self-repair declines while self-destruction rises. The germline releases the ratchet (elimination + de novo rebuild); this article asks whether somatic cells can, without the meiotic price. CEDAR (Centriolar Entropy-Damage Accumulation Ratchet) unifies a division counter with a state-lock mechanism: centriole loss releases identity in the direction set by the starting state (Renzova 2018: pluripotent → down; this protocol: differentiated → plastic → re-locked younger). The falsifiable protocol — controlled centriole elimination followed by verified de novo biogenesis, tested against OSK in a pre-registered 2×2 factorial — addresses every known barrier: the 53BP1-USP28-p53 arrest (targeted USP25/28 override), the PIDDosome-p53 axis (PIDD1-KD arm), acentriolar-mitosis and cGAS-STING (cytostatic window), genomic integrity (five-layer transformation surveillance), and cytoplasmic aging (SILAC purity proof and PTM-reset module). Eighty-eight references are verified via NCBI E-utilities.

1. The Problem and the Theory

1.1 The structural gap in aging research

The hallmarks of aging (López-Otín et al., 2013; 2023) enumerate twelve categories of age-associated damage. Most proposed interventions act on information carriers

— the epigenome (OSK reprogramming), telomeres, mitochondria — or on downstream phenotypes (proteostasis, autophagy). What is missing from this catalogue is a **persistent structural component**: an organelle that is semi-conservatively duplicated, retained across many divisions, and never reset in the somatic lineage.

The centriole is such a component. It duplicates once per cell cycle, templating daughter geometry from the mother's proximal surface; it is inherited asymmetrically in stem-cell divisions (Yamashita 2007; Wang 2009; Januschke 2011); and — unlike the epigenome — its structural state is not erased by transcription-factor-mediated reprogramming. The germline is the only compartment that clears it: centrioles are eliminated during oogenesis and rebuilt de novo in the early embryo (Kalbfuss & Gönczy 2023; Mikeladze-Dvali 2012). This monograph asks whether the somatic lineage can reproduce that clearing step — and what would happen if it did.

1.2 CEDAR — Centriolar Entropy-Damage Accumulation Ratchet

One sentence: the centriole is a structural ratchet that accumulates entropy-damage with each division (counter mode) and locks the cell's current differentiation state (state-lock mode); its loss releases identity in the direction set by the starting state.

- **Counter mode (CEDAR- α/β).** Semi-conservative duplication transmits templated geometry change (mother→daughter) plus superimposed stochastic errors: $\beta \cdot t + \eta(t)$, inseparable in vivo. The oldest centriole retained by the stem daughter accumulates the most temporal entropy.
- **State-lock mode (CEDAR- γ).** The Centrosome-Associated Memory Complex (CAMC) on the mother centriole acts as the ratchet pawl: it anchors the current differentiation state. Centriole loss releases the pawl; de novo assembly installs a fresh pawl that locks the new state.
- **The two-mode unification explains both counter-examples and the reprogramming barrier as one mechanism.** Renzova 2018 (loss in pluripotent cells → differentiation) is the ratchet released in the down direction; this protocol (loss in somatic cells → plasticity → re-locked younger) is the ratchet released in the up direction. The direction is set by the starting state, not by the perturbation.

Falsifiability. The ratchet model is falsified independently of any CRCS result if either (a) centriole loss in somatic cells produces the same outcome as in pluripotent cells (differentiation) despite p53 override and fibroblast identity — i.e., the starting state does not set the outcome; or (b) after verified de novo assembly the cell does not stably maintain the new state (rollback within 10 passages). Both are registered before data collection.

1.3 MCARA — Multi-Counter Architecture of Replicative Aging

The replicative limit is a **race of several counters**: centriole/CEDAR (#1), telomeres, mitochondria/ROS, epigenome. “Which counter’s entropy finishes first?” determines the phenotype of the limit. The 2×2 factorial of the protocol (OSK × reset) plus the hTERT control directly tests single- vs. multi-counter decomposition: if a single-counter reset (centriole) improves but does not abolish the limit, another counter finished first — and that is a testable, publishable result.

1.4 The germline as existence proof

The germline eliminates centrioles (meiosis/oogenesis) and rebuilds them de novo (blastomeres), clearing both programmed geometry and stochastic damage in one act — **without** any claim about information carriers. This is the existence proof that multi-counter entropy reversal is biologically possible. The Entropy Reset Protocol reproduces the centriolar step of this reversal in somatic cells **without the meiotic price**: diploidy preserved, no meiotic recombination, normal somatic DNA-repair capacity retained.

1.5 Placement of the claims

CEDAR (general theory): differentiation → division limit in animal stem cells
 > MCARA (special case): which counter's entropy finishes first
 > Entropy Reset Protocol (somatic reversal without the meiotic price)

PART 0 — THE RED THREAD (one-page mechanism)

THE RED THREAD — The Proximate Mechanism of Aging

One-page summary · 2026-08-11 · Jaba Tqemaladze, MD (Georgia Longevity Alliance)

The mechanism in one sentence

The organism ages because the oldest centrioles — carrying the most errors — accumulate in stem cells: division tempo falls and niche-signal perception falls (centrosome + cilium function), so tissue self-

repair declines while self-destruction rises — imbalance: repair↓ / destruction↑.

The chain (5 links)

1. **Retention [E]** — In asymmetric stem-cell divisions, the older (mother) centrosome is inherited by the cell that keeps stemness (Yamashita 2007, PMID 17255513; Wang 2009, PMID 19829375). Progeny do **not** eliminate centrioles (unlike *C. elegans*, where 88% of somatic cells eliminate; planarian neoblasts lack centrioles). The ratchet is never released in the somatic stem-cell lineage.
2. **Temporal entropy accumulation [H]** — Templated geometry change (mother→daughter, semi-conservative duplication; Kochanski & Borisy 1990, PMID 2335566) is **hypothesized to be PROGRAMMED — possibly a part of the differentiation mechanism itself** ($H_{\text{programmed}}$, falsifiable via cryo-ET comparison of de novo vs. templated geometry), and is inseparable from superimposed stochastic errors ($\beta \cdot t + \eta(t)$, not decomposable in vivo). Together they form the **temporal entropy** of the retained oldest centriole, which rises monotonically in daughter stem cells.
 - **v8.1 clarification (“contradiction 2”)**: “programmed” and “stochastic” are NOT contradictory for the same parameter — they are **two additive components acting at different levels**: $\beta \cdot t$ is a deterministic, systematic trend component (templated geometry drift, reproducible across cells), while $\eta(t)$ is the zero-mean stochastic fluctuation around it (cell-to-cell noise). This is the standard signal-plus-noise decomposition; the two are separable only in principle (in vivo they convolve, hence the cryo-ET falsification strategy compares de novo vs. templated geometry distributions, not single cells).
3. **Functional decline [I]** — The oldest centriole impairs two functions:
 - **Centrosome** → division tempo ↓ (defective spindle/duplication)
 - **Cilium** → niche-signal perception ↓ (antenna for Hedgehog/Wnt/PDGF α)
4. **Tissue imbalance [H]** — Division tempo ↓ + niche responses ↓ → **tissue self-repair** ↓, while **self-destruction** ↑ (apoptosis, senescence, degradation). Aging is the imbalance itself — not a “breakdown,” not a teleological “price.”
5. **Outcome** — Regeneration cannot keep up with destruction → stem-cell exhaustion, division limit, organismal aging.

The germline contrast

The germline is the only compartment that **releases the ratchet**: it eliminates centrioles (meiosis) and rebuilds de novo (blastomeres), clearing programmed geometry AND stochastic damage **in one act** — the existence proof that multi-counter entropy reversal is biologically possible. The **Entropy Reset Protocol**

reproduces this in somatic cells **without the meiotic price** (diploidy preserved, no meiotic recombination).

Placement (one hypothesis, one path, three levels)

CEDAR (general theory): differentiation → division limit in animal stem cells

- ▷ MCARA (special case): which counter's entropy finishes first
- ▷ Entropy Reset Protocol (somatic reversal without the meiotic price)

Where the thread runs (all documents)

Document	Version
CEDAR CONCEPT (LC/MCARA/CEDAR)	v1.8
MCARA CONCEPT (LC/MCARA)	v5.8
ENTROPY_RESET_PROTOCOL (ARGUS-OS3 / Desktop / GitHub)	v6.1
ENTROPY_RESET_GRANT_CORE (Marketing/docs / Desktop)	v2.0
ERC Part B1 (Marketing/ERC_StG_2026_CentrioleReset)	v4.5
ERC CONCEPT	updated
Entropy-in-Aging CONCEPT (Cologne workshop)	v1.4

GitHub: github.com/Georgia-Longevity-Alliance/ARGUS-OS3 · github.com/djabbat/LC · github.com/djabbat/Marketing

PART 1 — ENTROPY RESET PROTOCOL v8.3

Centriole Reset: Testing Structural Organelle Rejuvenation in Somatic Cells

Centriole Reset: Testing Structural Organelle Rejuvenation in Somatic Cells

Program name (grant platform): ENTROPY RESET PROGRAM **Version:** 8.3 | **Date:** 2026-08-12 | **Author:** Jaba Tqemaladze, MD (Georgia Longevity Alliance)

Scope declaration. This is a falsifiable research program, not a claim of established fact. It proposes an organelle-reset paradigm: controlled centriole elimination followed by verified de novo biogenesis, tested against partial epigenetic reprogramming (OSK), with genomic integrity as a non-negotiable

gate. Every statement is marked **[E] established** / **[I] inference** / **[H] hypothesis** / **[P] prediction**. Four levels of aging are distinguished throughout: **structural** (organelle), **cellular**, **tissue**, and **organismal** — no level is assumed from the success of the previous one.

1. Focused Research Question

[P] Can experimentally induced centriole elimination followed by verified de novo centriole biogenesis produce measurable, OSK-independent improvements in defined cellular aging phenotypes, while preserving genomic integrity?

This is deliberately narrower than “reset aging” — it is testable in a dish, pre-registrable, and falsifiable. **Explicit extrapolation boundary (per earlier review):** Phase 1 uses untransformed fibroblasts, which do **not** undergo asymmetric centrosome inheritance — Phase 1 tests feasibility of the reset (H1) and the cellular-aging phenotype (H2), **not** the stem-cell retention mechanism itself (which requires Phase 3 systems with demonstrated asymmetric inheritance, e.g., organoids or in vivo models). The red-thread mechanism is the motivation and the long-term target; Phase 1 does not claim to test it directly.

2. What Is Established [E]

2.1 Programmed centriole elimination is real

- Kalbfuss & Gönczy (2023, Open Biology, PMID 37963546): review of centriole elimination pathways.
- Kalbfuss & Gönczy (2023, Science Advances, PMID 37256957): **primary work** — 88% of *C. elegans* embryonic cells eliminate centrioles on a stereotyped schedule.

2.2 De novo centriole biogenesis is real

- Khodjakov et al. (2002, PMID 12356862): vertebrate cells assemble centrosomes de novo after removal.
- Takumi & Kitagawa (2022, Front Cell Dev Biol, PMID 35445021): experimental and natural induction of de novo centriole formation.
- Gönczy (2026, Nat Rev Mol Cell Biol, PMID 41310006): canonical biogenesis pathway (PLK4–STIL–SAS-6).
- **Aggregate-seeded de novo model (v7.4):** CEP152–CEP63–PCNT aggregates propagate outside the centriole and seed de novo biogenesis (Ozaki et al., 2025, bioRxiv, PMID 40667363, preprint — disclosed as such) — a molecular description of the template-independent route this protocol must prove.

- **Caveat [E]:** in most proliferating cells, the dominant route is canonical (next to the existing centriole); the de novo pathway is a specialized state — the protocol must prove it was used (see §7.3).
- **Centriole stability [E, v7.4]:** mechanisms underlying centriole stability are now systematically reviewed (Biven & Wang 2025, J Biol Chem, PMID 41167311) — the molecular basis for the “templated geometry change” claim (H_programmed) and its falsifiable cryo-ET test (§4.3).

2.2a Centriolar proteins and cellular senescence [E→I, v7.4]

- **POC5 regulates human adipogenesis and cellular senescence** (Pistorio et al., 2026, FASEB J, PMID 42507085) — a centriolar protein directly modulates a senescence-associated phenotype in primary human cells; independent support (beyond Robichaud 2024) for the centriole-senescence connection at the protein level.

2.2b The scaffold-stability argument (rebuttal)

Reviewer v11: “The centriole is a scaffold, not a hard disk; centrosomal proteins turn over rapidly, so PTM errors cannot accumulate.”

Rebuttal (with literature): (1) the centriole is a **semi-stable, long-lived organelle** — its structural proteins (notably centriolar tubulin) are not in rapid exchange with the cytoplasmic pool; the mechanisms maintaining this stability are now systematically reviewed (Biven & Wang 2025, J Biol Chem, PMID 41167311 — “Mechanisms underlying centriole stability”); (2) the tubulin PTM code (polyglutamylation, detyrosination, $\Delta 2$ -tubulin) is deposited **processively and irreversibly** on assembled microtubule arrays — the relevant enzymes (TTLL5/6, CCP5) act on polymerized tubulin, and $\Delta 2$ -tubulin is a terminal, non-removable modification (recent code reviews: J Biomed Sci 2026, PMID 42083040; Adv Sci 2025, PMID 40433930); (3) the centriole therefore accumulates PTM entropy **because** its subunits are retained — this is precisely the CEDAR premise, now anchored in the stability literature. Empirical resolution is pre-registered: PTM mass spectrometry on aged vs. de novo centrioles (Proof D, §6.3) directly measures whether accumulation occurs on the retained scaffold.

2.3 Centriole fate at fertilization is taxon-specific [E]

- Developmental and tissue-specific roles of mammalian centrosomes are now comprehensively reviewed (Meyer-Gerards & Bazzi 2025, FEBS J, PMID 38935637) — consolidates the [E] claims in §2.3–2.4 and the tissue-specificity caveat (Januschke 2011 neuroblast counterexample).
- Mouse oocytes lose centrioles during oogenesis (Simerly 2018, PMID 30143724; Schatten 2005, PMID 15385423).

- Starfish eliminate mother and daughter centrioles by distinct mechanisms (Borrego-Pinto 2016, PMID 27002173); *Drosophila* via Polo/ANA1 (Pimenta-Marques 2016, PMID 27229142).
- Avidor-Reiss & Fishman (2019, Reproduction, PMID 30496124): zygotes obtain two centrioles by diverse routes — paternal structures, maternal retention, or de novo.
- **Correction of earlier wording [v3.0→v4.0]:** there is **no universal “sperm centriole = seed, not a template.”** A defensible statement is: a paternal centriole or centriole-derived structure can provide a platform for subsequent centriole assembly in some taxa, whereas other metazoan systems rely substantially on de novo centriole formation.

2.4 Asymmetric centrosome inheritance biases fate [E]

- Yamashita et al. (2007, Science, PMID 17255513); Wang et al. (2009, Nature, PMID 19829375); Januschke et al. (2011, PMID 21407209); Barandun et al. (2025, PMID 39764850); Zhao et al. (2025, PMID 41315244); Chen & Yamashita (2021, PMID 33435817).
- **Centrosome heterogeneity in stem cells regulates cell diversity (v7.6):** Camargo Ortega & Götz (2022, Trends Cell Biol, PMID 35750615) — consensus review on centrosome heterogeneity in stem cells; Bolkent (2024, Genes to Cells, PMID 39379096) — mechanisms of asymmetric stem-cell division in tissue homeostasis. Both strengthen [E] status of the inheritance claim; an independent assessment of “centrosomes regulate fate via MT organization and signaling (Hedgehog/Wnt)” is consistent with, not contradictory to, CEDAR (cilium-mediated route H3/H8).
- **Cilium-aging axis (v7.5):** primary cilia shape the hallmarks of health and aging (Trends Mol Med review, 2023, PMID 37137787); a primary-cilia-autophagy axis in hippocampal neurons is essential to maintain cognitive resilience in aging (Nature Aging, 2025, PMID 39984747 — note: 2025, not 2024 as previously cited). Since the primary cilium is templated on the mother centriole, these findings independently strengthen the cilium-mediated route (H3, H8) — the cilium is the functional readout of centriole age.

2.5 OSK partial reprogramming is an established organism-level benchmark [E]

- Ocampo et al. (2016, Cell, PMID 27984723); Lu et al. (2020, Nature, PMID 33268865); Sahu et al. (2024, PMID 39259812); Zhang et al. (2026, PMID 41443352); Zielke & Ryan (2026, PMID 41856040).
- **Level-of-evidence asymmetry (critical assessment):** OSK is an **organism-level, multiply replicated benchmark** (progeria, vision, lifespan in vivo); centriole reset is a **cell-level, unreplicated hypothesis**. The 2×2 factorial compares them by design, but they are **never presented as evidence-equivalent**: OSK is the reference, reset is the candidate.

2.6 Germline remodeling vs. Hallmarks — precise wording [E→I]

Gametogenesis, fertilization and early embryogenesis are accompanied by extensive molecular and cellular remodeling that **overlaps conceptually** with several processes implicated in aging; **whether these processes constitute a coordinated rejuvenation program across the 12 Hallmarks remains unresolved** (López-Otín et al., 2013, PMID 23746838; 2023, PMID 36599349).

3. The Causal Gap, Prior Direct Evidence, and the Causal Evidence Map

The literature establishes: **centriole age → asymmetric inheritance / cell behavior**. It does **not** establish: **centriole age → organismal aging**, nor **centriole replacement → rejuvenation**.

3.1 Prior direct evidence — honest dialogue (evidence-based response)

The claim “no study has yet tested...” is true for the specific combination (aging-phenotype readouts + verified de novo origin + OSK factorial), but two near-neighbor studies must be addressed explicitly, and both cut against simple expectations:

Wong et al. (2015, Science, PMID 25931445). Centrinone reversibly depletes centrioles. **Critically, the original abstract states: centrosome loss irreversibly arrested normal cells in a senescence-like G1 state by a p53-dependent mechanism; only cancer cell lines proliferated indefinitely after loss, and upon washout each cancer line returned to an intrinsic centrosome-number set point.** [E] Thus the “elimination → recovery” cycle has been demonstrated **only in transformed (p53-compromised) lines**; in normal cells the outcome was irreversible p53-dependent arrest. This is a **central feasibility risk for Phase 1 of this protocol** (which uses untransformed fibroblasts): recovery after centriole loss in normal somatic cells is **an open empirical question to be demonstrated in Phase 1, not an established premise**. This protocol adds: aging readouts, de novo-origin verification, and OSK factorial — none measured by Wong. Novelty is therefore not the cycle, but the aging-oriented, origin-verified, factorial evaluation **in untransformed cells, where the cycle is not yet established**.

Tissue specificity of asymmetric inheritance (critical assessment). The claim “the ratchet is never released in the somatic stem-cell lineage” is **not universal even within one organism**: in *Drosophila* GSC, the mother centrosome is retained by the stem cell (Yamashita 2007); in *Drosophila* **neuroblasts, the daughter centrosome is retained by the stem cell and the mother centrosome goes**

to the differentiating GMC (Januschke et al., 2011, Nature Communications, PMID 21407209; Salzmann et al., 2014). The direction is **tissue-specific, not universal**. This protocol’s Phase 1 uses fibroblasts — a non-polarized, non-niche somatic cell in which the GSC-like vs. neuroblast-like classification is **not applicable by definition** (fibroblasts do not undergo asymmetric centrosome inheritance). The retention mechanism (§4.2a-6) is therefore presented as **cell-type-contingent**, and the ratchet claim is restricted to systems with demonstrated asymmetric inheritance. Counterexamples are registered, not omitted.

Renzova et al. (2018, Stem Cell Reports, PMID 30197118). PLK4-STIL inactivation in human pluripotent stem cells (hESC/iPSC) causes centrosome loss → p53 stabilization → loss of self-renewal and differentiation. [E] This is the closest published experiment, and its result **prima facie opposes H2** in pluripotent cells. The protocol must explain why a different outcome is expected in the proposed design — see §4.2. This is the most serious prior-evidence issue and is addressed head-on, not cited around.

3.2 Systematic evidence map (evidence synthesis, not quantitative meta-analysis — no comparable trials exist)

Causal link	Status
A. Centriole elimination is programmable	Established
B. De novo biogenesis is inducible	Established
C. Asymmetric centrosome inheritance affects stem-cell behavior	Supported
D. Centrosomal region participates in senescence signaling	Indirect (Robichaud 2024, PMID 39266565)
E. Centriole over-elongation occurs in pathology	Established but disease-specific (Köhrer 2023, Leukemia, PMID 37821581 — plasma cell disorders; whether analogous changes accumulate in physiological aging remains undetermined)
F. Centriole damage → biological aging	Speculative (CEDAR hypothesis)
G. Centriole replacement → rejuvenation	Untested
H. Centriole reset → multiple hallmarks simultaneously	Untested
I. Centriole reset → organismal rejuvenation	Untested

3.2b Quantitative meta-analysis plan (per earlier review)

Where quantitatively comparable outcomes exist across prototypes, a formal meta-analysis is planned (not just an evidence map):

- **Outcome:** centrosome-loss checkpoint passage/arrest rate (the p53/PIDDosome arrest fraction) — comparable across Uetake 2007 (RPE1), Mikule 2007 (RPE1 + tumor lines), Bazzi 2014 (mouse embryo), Wong 2015 (normal vs. cancer lines).
- **Estimator:** random-effects (DerSimonian-Laird) proportion meta-analysis; I² heterogeneity; forest plot.
- **Limitation (pre-specified):** n = 4–6 studies is small; the analysis is **descriptive of heterogeneity**, not confirmatory; individual-study definitions of “arrest” differ (G1 arrest vs. apoptosis vs. senescence-like) — subgrouping by assay.
- **Status:** data extraction in progress at the time of writing; the analysis is deposited with the SAP (OSF) before any Phase 1 data.

3.3 Novelty statement (per review — dialogue, not “first”)

To our knowledge, no study has yet directly tested whether controlled centriole elimination followed by bona fide de novo biogenesis — with four independent proofs of de novo origin — produces OSK-independent improvements in defined cellular aging phenotypes (CRCS) in somatic cells. Wong 2015 restored proliferation without aging readouts; Renzova 2018 showed differentiation upon acute loss in pluripotent cells (discussed in §4.2). The present program bridges the gap between these two results. (Systematic search re-verified before submission: centriole elimination AND rejuvenation → 0 hits; centrinone AND rejuvenation → 0; de novo centriole AND rejuvenation → 0.)

3.4 Extended review: all “elimination → recovery” experiments (critical assessment)

Systematic search (PubMed, 2026-08-11) identified **12 experiments** that eliminated centrosomes/centrioles and examined recovery. They divide into four families:

Family 1 — Physical ablation (laser/microsurgery): the de novo pathway was defined here

Study	System	What happened after elimination	Recovery outcome
Khodja kov et al. 200 2 (JCB 158:117 1–1181, PMID 123568 62)	BSC-1 (monkey fibroblasts) , laser ablation in S-phase	Acentriolar cells arrest; some assemble de novo centrosomes (γ-tubulin, pericentrin) from cytoplasm	De novo assembly possible; one surviving centrosome suppresses the pathway — complete removal required

Study	System	What happened after elimination	Recovery outcome
La Terra et al. 2005 (JCB 168:713-722, PMID 15738265)	HeLa, microsurgical removal in S	Cells progress through G1→S; de novo assembly in a subset	Pathway is “derepressed” only in fully acentriolar cells; recovery restores normal cycling
Uetake et al. 2007 (JCB 176:173-182, PMID 17227892)	Untransformed RPE1	p53/p21-dependent G1 arrest after centrosome removal; a subset assembles de novo	Recovery is inefficient in normal cells (most arrest); p53-deficient lines (U2OS) do not arrest
Mikule et al. 2007 (Nat Cell Biol 9:160-170, PMID 17330329)	RPE1 + tumor lines	Loss/integrity disruption → p38 → p53 → p21 → G1-S arrest	p53-defective tumor cells skip arrest → aneuploidy accumulation

Family 2 — Chemical depletion (reversible): centrinone

Study	System	Result	Recovery outcome
Wong et al. 2015 (Science 348:1155-1160, PMID 25931445)	RPE1, BJ + cancer lines	Centrinone (PLK4 inhibitor) depletes centrioles in ~3 days	In normal cells: irreversible senescence-like p53-dependent G1 arrest (NO recovery). In cancer lines: proliferation continues; washout returns each line to its intrinsic centriole set point. Recovery in untransformed cells is not established — it is the Phase 1 question of this protocol; no aging readouts were measured in either case
Renzova et al. 2018 (Stem Cell Reports 11:959-972, PMID 30197118)	hESC/iPSC	PLK4-STIL inactivation → centrosome loss → p53 stabilization → loss of self-renewal → differentiation into 3 germ layers	No recovery intended — chronic loss; outcome opposes H2 in pluripotent cells (resolved in §4.2)

Family 3 — Genetic models (developmental): the p53-response is universal

Study	System	Result	Recovery outcome
Bazzi & Anderson 2014 (PNAS 111:E1491-E1500, PMID 24706806)	Mouse embryo, SAS-6 KO (acentriolar)	Acentriolar mitosis in early embryo → p53-dependent apoptosis ; some cells divide via acentriolar spindles	No recovery — lethality; proves p53 is the gatekeeper of the acentriolar state in vivo
Xie et al. 2021 (Dev Cell 56:3334-3348, PMID 34932949)	Zebrafish/mouse endoderm	Centrioles needed to restrain p53-mediated apoptosis when ERK signaling is absent	Context-dependent survival; reinforces p53 as the central gate
Mikeladze-Dvali et al. 2012 (Development 139:1670-1679, PMID 22492357)	C. elegans oogenesis	Natural programmed elimination of all 4 daughter centrioles in oocytes	No somatic recovery — embryo reacquires centrioles from sperm (germline reset)

Family 4 — Natural de novo in parthenogenesis (proof de novo suffices for development)

Study	System	Result
Szöllosi & Ozil 1991 (Biol Cell 72:61-66, PMID 1756312)	Rabbit embryos, parthenogenetic activation (no sperm)	De novo centrioles form in blastomeres — development proceeds without any sperm-derived centriole
Riparbelli et al. 2017 (Results Probl Cell Differ, PMID 28779329)	Insect parthenogenesis	“Centriole renaissance” — de novo assembly supports full development in several insects
Inoue et al. 2018 (BioEssays 40:e1700135, PMID 29522658)	Review — vertebrate MTOC	Maternal centrosomal proteins rebuild the MTOC after the germline elimination — “loss and rebirth”

Family 5 — New chemical tools (2025-2026): toward cleaner depletion

Study	Tool	Relevance
Jeong et al. 2025 (J Med Chem, PMID 41329867)	Aminopyrazole PLK4 inhibitors	Next-generation centrinone-class tools, higher selectivity

Study	Tool	Relevance
PLK4-targeting reviews 2025 (v7.4)	PLK4 as therapeutic target: breast cancer (Parsyan 2025, Carcinogenesis, PMID 41092110), skin cancer (Jaiswal 2025, Cells, PMID 40940791), general oncology (Muntaqua 2025, iScience, PMID 41488365)	Clinical pipeline context — validates PLK4 as druggable axis; informs the safety argument (§6.5b: PLK4 inhibition is not itself transforming when transient)
PLK4-PROTAC (PMID 37279162, J Med Chem 2023)	First potent, selective, in vivo efficacious PLK4 proteolysis-targeting chimera	Fast, clean, catalytic depletion — no sustained kinase-occupancy artifacts; see §6.2b
AID2 degron on centrosomal proteins (PMID 40020058, Sci Adv 2025)	Auxin-inducible degron 2 (TIR1) degrades essential centrosome protein CEP192 in live mice	Proof that AID2 achieves rapid, reversible, sustained degradation of centrosomal proteins in vivo — the template for SAS-6/PLK4 degron lines
Lu et al. 2025 (Cell Reports, PMID 40974574)	MCIDAS nuclear-cytoplasmic translocation	Massive de novo centriole amplification (multiciliogenesis) — the natural “boost” mechanism
FBXW7-mediated endogenous degradation Plk4 (PMID 41453690, JBC 2026)	E3 ligase degrades hyper-phosphorylated Plk4 — the cell’s own copy-number brake	Perspective for genetic modulation of elimination kinetics (not a chemical tool)

Synthesis for this protocol (evidence-based response): (1) the p53/p21 arrest after centrosome loss is **universal** (Khodjakov, La Terra, Uetake, Mikule, Bazzi, Xie) — justifying a **targeted p53-override** as a mandatory component; **v7.0 precision upgrade** : the arrest is mediated by the **mitotic surveillance pathway 53BP1-USP28-p53** (Fong 2016, eLife; Meitingner 2016, JCB; Wang 2021, EMBO J — in vivo microcephaly), NOT the classical DNA-damage response — therefore global pifithrin- α is replaced by **selective USP25/28 inhibition** (Bratt 2025, Cell Chem Biol; Hernandez-Olmos 2026, J Med Chem), which unlatches only the USP28 bridge and **preserves the p53 response to genuine DNA damage**, protecting the Genome-integrity gate (H5); pifithrin- α is retained only as the arm-E comparator (see §6.1); **v7.3 addition** : the **complementary PIDDosome arm is registered** — supernumerary/aged centrosomes activate the **PIDDosome-p53 axis via centriolar distal appendages/ANKRD26** (Fava 2017, Genes Dev; 2021 EMBO J, PMID 33350486) and extra centrosomes induce **PIDD1-mediated inflammation** (2023 EMBO J, PMID 37530438); the protocol therefore monitors **both directions** of the centrosome-p53 surveillance axis — loss (53BP1-USP28) and amplification (PIDDosome/ANKRD26) — with the centriole-number gate E (exactly 2) as the switch that prevents either arm from firing post-recovery; (2) recovery of division after centrinone washout is **proven only in transformed lines** (Wong 2015 — corrected in v5.6) and aging phenotypes were never measured — our CRCS closes exactly that

gap; (3) de novo assembly is **inefficient/error-prone in untransformed cells** (Uetake 2007; La Terra 2005 — initial numbers uncontrolled) — motivating the PLK4-pulse boost, number titration, and the 4-proof verification; (4) parthenogenesis proves de novo suffices for organismal development (Szöllosi 1991; Riparbelli 2017) — the strongest existence proof for the paradigm.

3.5 The prototype chain and the actual novelty (2026-08-11,)

The four nearest prototypes form an ascending ladder — each solved one technical piece, and each left the same gap:

Prototype	Solved	Left open
Khodjakov et al. 2002 (CHO, laser microsurgery)	Centriole → removed → new structure built from zero	Cell-cycle/aging phenotype after rebuild
La Terra et al. 2005 (HeLa, laser/needle)	Resident centrioles removed → multiple new centrioles de novo	Number control (initially uncontrolled); aging phenotype
Uetake et al. 2007 (normal human cells)	De novo assembly after removal is a general capacity of mammalian somatic cells , not a cancer-cell curiosity	Efficiency low; p53 arrest; aging phenotype
Wong et al. 2015 (centrinone)	Chemical path: deplete → wait → restore PLK4 → de novo rebuild (transformed lines)	Normal-cell recovery (irreversible p53 arrest); aging phenotype

The actual novelty, stated strictly: removing the old centriole is not new. De novo creation of a new centriole is not new. **What is new is the question:** does the aging state of the old somatic cell change as a result? More rigorously: **old cell + old centriole → old cell + genuinely new centriole** — then ask: did the cellular age phenotype change? Every prototype stopped at “the centriole came back”; none measured whether **the cell came back younger**. That single missing measurement is this protocol.

Novelty as binding, not as “first” (, 15:36): “nobody has done anything like this” would be wrong — **more than half of the technological experiments have already been done** (Khodjakov, La Terra, Uetake, Wong solved the pieces). The real novelty is the **binding**: assembling the existing pieces into an **aging causal test** — elimination + verified de novo + aging-phenotype readouts + OSK factorial + safety gate — none of the prototypes performed the binding. The novelty claim is therefore about the integration into a causal test, explicitly not about the individual technologies.

Visual (13 puzzle pieces — schematic only, v8.0): [docs/ENTROPY_RESET_13_PUZZLES.png](#) — 12 Hallmarks of Aging + centriole elimination

& de novo synthesis assembling into one picture. **this is explicitly labeled a schematic for grant/lecture use, NOT a claim that the centriole is a 13th hallmark** — in the publication it moves to supplementary material with the caption “schematic; the centriole is a candidate mechanistic axis, not an additional hallmark.”

To our knowledge, no study has yet directly tested whether controlled centriole elimination followed by bona fide de novo biogenesis — with four independent proofs of de novo origin — produces OSK-independent improvements in defined cellular aging phenotypes (CRCS) in somatic cells. Wong 2015 restored proliferation without aging readouts; Renzova 2018 showed differentiation upon acute loss in pluripotent cells (discussed in §4.2). The present program bridges the gap between these two results. (Systematic search re-verified before submission.)

4. Hypotheses (pre-registered, falsifiable)

ID	Hypothesis	Falsified if
H1 (Mechanistic)	Controlled elimination + bona fide de novo biogenesis is feasible in untransformed somatic cells	De novo origin cannot be demonstrated (remnant regeneration instead); or recovery is blocked by p53-dependent arrest (technical barrier, see §2.1 Wong) — this falsifies H1-as-feasibility, NOT H2
H2 (Functional)	Cells with verified de novo centrioles improve defined cellular-aging phenotypes vs. matched controls	$B \leq A$ on primary composite (CRCS)
H3 (Independence)	Centriole reset has an effect detectable after accounting for OSK	No significant main effect of reset in 2×2; interaction absent
H4 (Causality)	Rejuvenation magnitude correlates with extent/fidelity of replacement, not with the perturbation used	Perturbation control (failed reset) shows equal effect
H5 (Safety)	Reset does not increase genomic instability, aneuploidy, or abnormal centrosome number beyond acceptance criteria	Genome-integrity gate fails in $\geq 5\%$ of replicates
H6 (p53-control)	Transient p53 suppression is a controlled variable, not a free technique : any CRCS effect is attributable to centriole reset only if it exceeds the effect of p53-suppression alone	p53-suppression-only arm (no reset) matches reset arm on CRCS

ID	Hypothesis	Falsified if
H6' (USP2 8-specificity , v7.0)	The targeted USP28-inhibitor override does not globally disable p53: cells pass the acentriolar window but retain a functional p53 response to independent DNA damage	Etoposide challenge (γ H2AX/p21 induction) fails in USP28-treated cells, OR checkpoint passage rate does not improve vs. pifithrin- α

4.2 Addressing the closest prior experiment: Renzova et al. 2018 (critical assessment)

Renzova et al. (2018, PMID 30197118) inactivated the PLK4–STIL module in human pluripotent stem cells; centrosome loss caused p53 stabilization, loss of self-renewal, and differentiation — prima facie the opposite of H2. Five design differences explain why a different outcome is expected here, and each is testable:

Difference	Renzova 2018	This protocol	Rationale
1. Cell type	hESC/iPSC (pluripotent)	Somatic fibroblasts	Pluripotency maintenance is centrosome-dependent (Renzova); fibroblast identity is not. “Centrosome loss breaks stemness” is irrelevant to a non-stem cell
2. p53 state	p53 intact differentiation	selective inhibition during the acentriolar window (v7.0; pifithrin-α as arm-E comparator)	The p53-dependent differentiation mechanism (Renzova) is blocked; the plasticity window opens without differentiation
3. Duration	Chronic loss (self-renewal impossible)	Acute elimination → verified de novo recovery to exactly 2 centrioles	The hypothesis concerns the post-recovery state, not the acentriolar state
4. Recovery	None intended	PLK4 pulse → functional centrioles (ultrastructure, ciliogenesis, faithful duplication)	H2 tests the rebuilt organelle, not loss itself
5. Readouts	Self-renewal markers	Aging phenotypes (CRCS), not stemness	Loss of self-renewal in Renzova is not a measure of cellular rejuvenation

[P] Prediction: in somatic fibroblasts with p53 suppressed during the window and centrioles rebuilt de novo to 2 per cell, the Renzova outcome (differentiation) should

not occur, while CRCS improvement may. If differentiation/self-renewal loss occurs anyway → H2 falsified (pre-registered). Renzova is treated as the **primary prior evidence to be overcome**, not ignored.

4.2a The differentiation-ratchet interpretation: why Renzova actually supports the paradigm [H]

Renzova 2018 is not merely a counter-example to be overcome — under the **centriole-as-differentiation-ratchet** hypothesis it is direct supporting evidence. The hypothesis: the centriole/centrosome is a **structural ratchet that locks the cell's current differentiation state**. It does not only count divisions; it anchors identity.

- **Direction of the ratchet:** stem/pluripotent → differentiated (normally irreversible; reprogramming is the rare exception).
- **Removing the ratchet (centriole loss) releases the anchor:**
 - In a **pluripotent cell** (Renzova 2018): the anchor of pluripotency is removed → the cell rolls down the ratchet → spontaneous differentiation into three germ layers. This is exactly what a ratchet model predicts: without the anchor, the pluripotent state cannot be maintained.
 - In a **somatic cell** (this protocol): the anchor of the differentiated state is removed → the cell is plastic (neither pushed down — it is already at the bottom — nor locked in place) → the plasticity window opens → a **new de novo centriole re-locks a new state** (reprogrammed/rejuvenated).
- **Molecular mechanism [H]:** CEDAR-γ — a **hypothetical centriolar state-locking mechanism** (placeholder term; formally “Centrosome-Associated Memory Complex” — see caveat below) on the mother centriole is the ratchet pawl: it maintains the differentiated state. Centriole loss releases the pawl; de novo assembly installs a fresh pawl that locks the new state.
- **CAMC caveat (v8.0):** a critical assessment noted that “Centrosome-Associated Memory Complex” is **not an established structure** in the literature. It is therefore renamed “**hypothetical centriolar state-locking mechanism**” in all submission text, and the molecular burden is placed on **specific candidate carriers** rather than an unobserved complex: (a) **distal appendages** (the ANKRD26-PIDD1 site — PIDDosome axis, Fava 2017, PMID 28130345; ANKRD26-PIDD1 recruitment, PMID 33350495) as a candidate architectural anchor of “age/identity”; (b) **PTM code on the mother centriole** (polyGlu, Δ2-tubulin) as the state-bearing substrate; (c) **CEP152/PCNT-containing PCM** as the signaling-competence layer. The state-lock hypothesis is retained, but its molecular implementation is now falsifiable per-candidate (each candidate has a perturbation arm), not tied to an unobserved complex.

CEDAR unified definition (v5.2): CEDAR — **Centriolar Entropy-Damage Accumulation Ratchet** — is one ratchet with two modes of action: **(1) the counter mode** (CEDAR-α/β): semi-conservative duplication accumulates

entropy/damage with each division ($\beta \cdot t$, $-\lambda \cdot N_{\text{asym}}$); **(2) the state-lock mode** (CEDAR- γ): the same organelle acts as a **differentiation ratchet** that locks the cell's current identity (pluripotent or differentiated). Loss of the ratchet releases identity in the direction set by the starting state (Renzova: pluripotent $\rightarrow$ down; this protocol: differentiated $\rightarrow$ plastic $\rightarrow$ re-locked younger). The two modes explain both the aging counter and the reprogramming barrier as one mechanism.

Related models (links only): MCARA (Multi-Counter Architecture of Replicative Aging) — replicative limit as a race of several counters, **with the centriole as counter #1 (the centriolar ratchet)**: ~/Desktop/LC/MCARA/CONCEPT.md (v5.8). CEDAR unified definition (centriolar counter + state-lock ratchet): ~/Desktop/LC/MCARA/CEDAR/CONCEPT.md (v1.8). - **Testable predictions that distinguish the ratchet model from a mere “stress” model:** (1) after de novo assembly, the new state must be **stably maintained** (CRCS improvements persist over serial passages — no rollback); (2) the same centriole loss in pluripotent cells causes differentiation (Renzova) while in somatic cells with p53 suppressed it causes plasticity — the direction of the outcome is set by the starting state, not by the perturbation; (3) reintroducing an aged centriole into a reset cell re-locks the old state (transplantation arm predicts rollback).

Synthesis: Renzova 2018 measured the ratchet's release in one direction (pluripotent $\rightarrow$ down). This protocol tests the same release in the other direction (differentiated $\rightarrow$ plastic $\rightarrow$ re-locked younger). The two results are opposite sides of one mechanism — the ratchet is unidirectional, so the outcome of centriole loss depends entirely on the starting state. This is the deepest available reconciliation of the closest prior experiment with the hypothesis.

Pre-specified falsification of the ratchet model alone (evidence-based response): the ratchet model is falsified **independently of H2** if either of two patterns occurs: (a) **no direction-dependence** — centriole loss in somatic cells produces the same outcome as in pluripotent cells (differentiation/loss of identity) despite p53 suppression and fibroblast identity, i.e., the starting state does not set the outcome; or (b) **no re-locking** — after verified de novo assembly the cell does not stably maintain the new state (rollback to the old phenotype within 10 passages) even though CRCS improves transiently. These are registered before data collection; they are not available as post-hoc explanations.

4.3 Mechanistic hypotheses (peer-review-critical: “through what pathway?”)

For each mechanistic hallmark, a concrete molecular route from centriole state to phenotype is proposed (all [H], each with a specific assay):

Hallmark	Mechanistic route [H]	Pathway assay
H3 Epigenome	The primary cilium is templated on the mother centriole; ciliary signaling (Hedgehog-GLI, Wnt) feeds back on chromatin regulators (PRC2, TET). A fresh mother centriole → fresh cilium → rebalanced GLI/TET activity → DNA-methylation drift partially reversible	Ciliary proteome MS; GLI1/2 activity; TET2 ChIP; EPIC clock
H1 Genome	Aged/supernumerary centrioles cause segregation errors (Shin 2021); a verified 2-centriole set restores error-free division. Note: accumulated SNVs are NOT erased — H1 is limited to segregation/structural errors	Micronucleus, γH2AX, scDNA-seq
H4 Proteostasis	PolyE on aged centriolar tubulin (TTLL5/6) seeds the nucleus-to-cilium scaffold (Robichaud 2024); a polyE-free centriole removes the seed. Insoluble-aggregate burden assayed independently	GT335; ProteoStat; insolubility assay
H5 Autophagy	Centriole loss induces selective autophagic clearance of centrosomal remnants; recovery normalizes flux. Measured, not assumed	LC3 flux, p62, mito-Keima
H7 Mitochondria	No direct route — indirect via ciliary signaling (Wnt/mTOR) and reduced SASP-driven ROS. Covariate, not causal claim	Seahorse; mitoSOX; heteroplasmy
H8 Senescence	Direct route: polyE MT scaffold → KIFC3 → FBF1 → PML (Robichaud 2024). Fresh centriole lacks the scaffold	SA-β-gal; p16/p21; SASP panel; KIFC3/FBF1 localization
H9 Stem cells	Asymmetric inheritance: mother centriole retained by stem cell (Yamashita 2007, Wang 2009). A symmetric young pair removes the “pre-aged” bias	Lineage tracing; centrosome tracking; serial passaging

Honest scope: routes marked direct (H4, H8, H9, H3) have a published molecular mediator, **but causality in the reset direction is untested** (correction: “direct route” → “plausible mediator, causal direction untested”); H1 is limited to segregation errors; H5, H7 are measured as covariates with no

causal claim. This converts the program from “measure everything, hope” to a set of testable pathway hypotheses.

Test of the “programmed geometry” claim (critical assessment): the claim “templated geometry change is programmed — part of the differentiation mechanism” is **not assumed; it is made falsifiable**. H_programmed: the geometry of de novo-assembled centrioles (cryo-ET: length, cartwheel integrity, over-elongation rate) **differs** from templated (aged) ones. Prediction: if de novo centrioles in the same cell type are geometrically indistinguishable from aged templated ones → geometry is a passive product of assembly, and the “programmed” claim is weakened; if they are systematically different (e.g., young-like length/cartwheel) → the programmed/developmentally-coupled interpretation is supported. Measured in Proof D (de novo functional verification), n = 20–30 cells/condition, cryo-ET.

5. Causal Architecture (levels of evidence ladder)

AGING → centriole structural state → centriole inheritance → stem-cell functional state
→ tissue regenerative capacity → age-associated phenotype

Intervention: centriole elimination → verified de novo biogenesis → functional young centriole
→ cellular phenotype → stem-cell phenotype
Positive control (parallel): OSK → epigenetic rejuvenation
Central experiment: OSK × centriole reset

Proof ladder — each level possible only on success of the previous: 1. **Organelle** — structure/age (EM, cryo-ET, PTM mass spec) 2. **Cell** — division, senescence, mitochondria, epigenome (CRCS) 3. **Stem cell** — self-renewal, differentiation, regenerative capacity 4. **Tissue** — organoid/regeneration 5. **Organism** — healthspan/lifespan (in vivo, only if 1–4 pass)

No systemic “rejuvenation” claim is made at levels 3–5 without data at those levels.

6. Experimental Design

6.1 The 2×2 factorial with formal interaction

	Centriole reset –	Centriole reset +
OSK –	A (control)	B
OSK +	C (OSK benchmark)	D (OSK + reset)

Statistical question (per independent verification): not “B > A” alone, but — is there a significant **main effect of centriole reset** and a significant **OSK × reset interaction** (two-way ANOVA / mixed model)? This separates an additive mechanism from an OSK-dependent one.

Temporal-dissociation arm F (v8.2): OSK cyclic expression takes 10–14 days (Ocampo 2016; Lu 2020); centriole elimination + de novo takes 3–5 days. In arm D (simultaneous OSK+Reset), OSK-driven cytoplasmic remodeling (chaperones, TTL5/6) could “pre-clean” the cytoplasm before de novo assembly — making H3 (independence) untestable. Therefore a **sixth arm F is added: Reset → washout → 10 passages → OSK**, testing whether the structural reset creates **long-term memory** that survives subsequent cytoplasmic/epigenetic aging. Prediction under CEDAR: reset benefits persist through the washout-and-aging interval (structural memory); prediction under the confounder: benefits decay as the cytoplasm re-ages (OSK-dependent artifact). Arm F is the temporal control that makes H3 interpretable; without it, arm D cannot separate “reset independent of OSK” from “OSK pre-cleaned the cytoplasm.”

p53-suppression confounder control (critical assessment): obtaining de novo centrioles in an untransformed cell likely requires **transient suppression of the p53/PIDDosome axis** (Wong 2015: normal cells arrest irreversibly without it). p53 suppression alone can alter hallmark readouts (genomic stability, senescence). Therefore a **dedicated control arm E** is added: p53-suppression only, no centriole reset, no OSK. H6 is falsified (and all reset claims are confounded) if arm E matches arm B on CRCS. This makes p53 the **controlled variable of design**, not an implicit technique.

Targeted override — arm E' (v7.0, revision): the p53 arrest after centrosome loss runs through the **mitotic surveillance pathway 53BP1-USP28-p53** (Fong 2016, eLife; Meitinger 2016, JCB), not the classical DNA-damage response. Global pifithrin- α (arm E) disables p53 transcription globally, risking silent genomic lesions that WGS detects too late. **Arm E' (primary override): selective USP25/28 inhibition** (Bratt 2025, Cell Chem Biol; Hernandez-Olmos 2026, J Med Chem; structural basis: Patzke 2024, EMBO Rep, PMID 38816515) — unlatches the USP28 bridge only, leaving p53 competent for genuine damage. **Genetic alternative (v7.5):** transient **CRISPRi against USP28 or 53BP1** (dCas9-KRAB, doxycycline-inducible) achieves the same targeted override without any chemical exposure — pre-registered as the orthogonal confirmation route (chemical vs. genetic override must agree on checkpoint passage rate). **Dual specificity control:** arms E (pifithrin- α -only) and E' (USP28-inhibitor-only) run in parallel, both without reset/OSK; H6' requires: (a) checkpoint passage rate $E' \geq E$ (override works), (b) etoposide-challenge p53 response preserved in E' but not E (specificity), (c) $E' \leq A$ on CRCS (no independent rejuvenation signal). If E' fails (a), the technical-barrier gate (§13 Phase 1) applies; if E' shows a CRCS signal matching B, all reset claims are confounded (H6 falsified).

6.1a OSK × reset interaction — interpretation matrix (critical assessment)

Pre-specified interpretation of the 2×2 factorial, **locked before data**:

Main effect of reset	OSK × reset interaction	Interpretation	Conclusion for H3
+	+ (synergy)	Reset amplifies OSK beyond additivity	H3 supported (independence of mechanism, super-additive benefit)
+	0 (additive)	Reset adds to OSK independently	H3 supported (independent additive mechanism)
+	– (sub-additive/ antagonistic)	Reset partially substitutes for OSK	H3 qualified: benefit exists but not “beyond OSK” — reframe as alternative route
0	0	Reset alone does nothing	H3 rejected
0	+ (only with OSK)	Reset requires OSK context	H3 rejected as “independent”; mechanism is OSK-gated

Rule: the interaction is interpreted **jointly with main effects** (never in isolation); the matrix is registered verbatim; deviation from the matrix requires a pre-registered amendment.

6.2 Centriolar perturbation control (the “not-a-reset” arm)

A fifth condition that induces maximal comparable cellular stress **without successful de novo reset** (e.g., PLK4 inhibition + recovery blocked by a biogenesis inhibitor, or abortive ablation). This discriminates: **removal → bona fide reset** from **removal → stress → survivor selection**. Any rejuvenation seen in the perturbation arm falsifies H4.

6.2b Targeted-degradation elimination route (v7.1, revision)

Rationale: chemical PLK4 inhibition (centrinone, aminopyrazole) acts by sustained kinase occupancy; **targeted degradation removes the protein catalytically** and is cleaner. Three validated routes exist (all confirmed via PubMed): - **AID2 (auxin-inducible degron 2):** TIR1-based, auxin-triggered degradation; already applied to the essential centrosome protein **CEP192 in live mice** (PMID 40020058, Sci Adv 2025) — the template for **SAS-6-mAID / PLK4-mAID** degron lines (elimination by dilutive loss over ~3 divisions, as in centrinone, but without kinase-occupancy off-targets). - **dTAG system:** FKBP12(F36V)-tagged SAS-6/PLK4 degraded by dTAG-13 (cell lines; knock-in protocol, Methods Enzymol 2023, PMID 36764753). - **PLK4-PROTAC:** first potent, selective, in vivo efficacious PLK4 chimera (PMID 37279162, J Med Chem 2023) — direct chemical degradation of PLK4.

Decision rule (pre-registered): Phase 1 tests **AID2-SAS-6** as primary elimination route (reversible, orthogonal to chemistry), with centrinone as the established comparator and PLK4-PROTAC as the alternative; the three routes must yield **concordant passage rates** before CRCS claims (route-concordance gate).

Dual-synchronization protocol (AID2 trap): AID2 degradation is not instantaneous (24–48 h, dilutive at division); cells already in G2 when auxin is added would enter mitosis with partially degraded centrioles → multipolar spindles → micronuclei → cGAS–STING (exactly the v7.9 trap). CDK4/6i alone does not stop G2 cells. Therefore the elimination protocol requires **two sequential blocks**: (1) **aphidicolin (S-phase block)** synchronizes the population; (2) after washout, **auxin + RO-3306 (CDK1i, strict G2 block) for 12 h** — all cells complete DNA replication but cannot enter mitosis until SAS-6 degradation is complete (verified by western blot, >90% loss). Only then is the CDK1 block released into the cytostatic window. This converts “eliminate while some cells are in G2” into “eliminate with the entire population pinned in G2.”

Resistance caveat (v7.4): mammalian cells can develop **resistance to auxin-inducible degradation** (Hyle et al., 2026, J Biol Chem, PMID 42248454 — mechanisms: degron mutation, TIR1 downregulation, pathway compensation). Therefore the route-concordance gate is not only a quality filter but a **resistance screen**: if AID2-SAS-6 fails while chemical routes succeed, resistance — not biology — is the explanation, and the elimination route is switched (pre-registered).

6.2c The acentriolar-mitosis trap: cytostatic window + cGAS/STING dissociation arm (v7.9)

Problem (fatal-flaw candidate, accepted): USP25/28 override prevents the primary 53BP1-USP28-p53 arrest, but a cell entering mitosis without centrioles forms an acentriolar spindle → lagging chromosomes → micronuclei → **micronucleus rupture activates cGAS-STING** (Mackenzie 2017, Nature, PMID 28738408; Dou 2017, Nature, PMID 28976970) → **secondary, USP28-independent senescence/SASP**. A bypass that forces acentriolar mitosis is therefore insufficient — it merely trades the primary arrest for a secondary inflammatory barrier.

Solution 1 — Cytostatic window (primary): during the elimination→rebuild window (48–72 h), an **reversible CDK1 inhibitor (RO-3306) or CDK4/6 inhibitor (palbociclib)** holds cells in G2/G1, preventing mitotic entry until exactly two functional centrioles are verified (gate E) and appendage maturation is underway. Mitosis resumes only post-verification. This converts the window from “survive mitosis without centrioles” to “never enter mitosis without centrioles” — eliminating the acentriolar-mitosis trap at its source.

Solution 2 — cGAS/STING dissociation arm (control): a parallel CRISPR-KO of **cGAS or STING1** runs the same elimination protocol without the cytostatic block. Predictions: (a) if survival/passage rate rises sharply in cGAS/STING-KO vs. parental under identical override, the **mitotic-error stress (not organelle loss itself) is**

the dominant barrier — reported as a mechanistic dissociation; (b) if KO does not rescue, cGAS-STING is not the limiting barrier and the cytostatic window is the necessary route. Either outcome is a pre-registered, interpretable result; the arm is not “remove the safety” but “localize the barrier.”

Solution 3 — PIDDosome bypass arm (v8.0): a critical assessment noted that USP28-i blocks only the 53BP1-USP28 bridge, while the **PIDDosome-p53 axis** (Fava 2017; ANKRD26-PIDD1 at distal appendages, PMID 33350495) can fire **independently** upon centrosome loss/amplification — and that the cytostatic window prevents mitosis but does not prevent G1 p53 activation via PIDD1. Therefore a **third bypass arm is added: transient PIDD1-KD or ANKRD26-KD (dCas9-KRAB, dox-inducible)** in Phase 1, running in parallel with USP28-i and the cytostatic window. Pre-registered comparison: checkpoint passage rates under (i) USP28-i alone, (ii) PIDD1-KD alone, (iii) both, (iv) both + cytostatic window. If PIDD1-KD is required (passage rate rises only when added), the PIDDosome is a dominant G1 barrier and the override strategy is PIDD1-targeted; if USP28-i alone suffices, PIDD1 is not rate-limiting in this system. Either outcome updates the override logic pre-registered — never post-hoc.

Safety note: the cytostatic window does NOT remove the genome-integrity gate (H5) or transformation surveillance (§6.5b) — it adds a timing control that prevents the very chromosome-segregation errors those gates would otherwise detect. The three solutions together bound the v14/v15 traps: no acentriolar mitosis (window), direct barrier localization (cGAS/STING arm), and a second, USP28-independent p53 gate (PIDDosome arm).

Clarification (peer-review rebuttal, v7.1): targeted degradation changes the **elimination route** — it does **not** remove the need for p53-override. Loss of SAS-6/PLK4 still leaves cells acentriolar → the **mitotic-surveillance p53 checkpoint still fires** (Fong 2016; Meitinger 2016). The USP25/28 override (v7.0) remains the **survival gate**; AID2/PROTAC is the **elimination gate**. The two are orthogonal and both mandatory.

6.3 De novo origin — four independent proofs (A-D)

Proof	Method	Excludes
A. Old centrioles truly gone	Multi-marker IF (CETN2, CEP152, SAS-6) + serial EM + live imaging	Marker-specific loss
B. New centriole exists	Dendra2-Centrin1 photoconversion; SNAP-tag pulse-chase	Residual signal
C. Genuinely de novo (no template)	Label-free detection of new assembly; CLEM correlative	Regeneration from remnant

Proof	Method	Excludes
C'. Molecular purity — SILAC (cytoplasmic-aging control)	SILAC pulse-chase mass spectrometry during the de novo window: heavy-isotope labeling (Lys-8/Arg-10 or R10K8) starts at elimination; the new centriole must be assembled predominantly from newly synthesized tubulin/centriolar proteins (heavy-labeled), not recycled pools from the old centriole (SILAC+centrinone combination already validated: PMID 32501498)	“Recycled old bricks” — a de novo centriole built from aged cytoplasmic building blocks would re-acquire old PTMs (oxidized tubulin, dysregulated TTL5/6 polyglutamylation) and the “reset” would be cosmetic
D. Functional	Ultrastructure (cryo-ET), molecular composition (MS), spindle function, ciliogenesis, faithful duplication	“Repair,” not reset
D''. Geometry — cryo-ET subtomogram (v7.7)	Subtomogram-averaged cryo-ET: triplet A-C tubule angles, cartwheel integrity, dense-ring presence, over-elongation rate ($n = 20\text{--}30$ cells/condition, per §13); the de novo centriole must match embryonic/young reference geometry , not aged templated geometry — this is the direct, quantitative test of H_programmed	Geometry is passive vs. programmed: if de novo geometry is indistinguishable from aged templated, the “programmed geometry” claim is weakened (pre-registered in §4.3)
D'. Maturation — appendages & ciliogenesis (v7.7)	De novo centrioles lack distal/subdistal appendages for several cell cycles (maturation: CEP295/Ana1 — Pimenta-Marques 2024, EMBO Rep, PMID 38200359; distal centriolar protein network — Wang 2018, Nat Commun, PMID 30258116); without subdistal appendages the centriole cannot anchor MTs or template a primary cilium. Proof D' adds: time-course of appendage acquisition (ODF2, Ninein, CEP295 IF + ultrastructure), first-cilium latency (ciliogenesis endpoint), and the criterion that a “functional young” centriole must be both new AND mature (appendages + ciliogenesis within a pre-registered window, e.g., ≤ 5 passages)	A “young” but immature centriole is functionally blind to niche signals — H3/H8 readouts would be confounded by maturation delay, not reset

Proof	Method	Excludes
D''''. Maturation boost ("Killer 2.0")	Optional inducible CEP295/Ana1 expression during the de novo window to accelerate appendage acquisition (CEP295 is the essential maturation factor — Pimenta-Marques 2024); predicts compression of first-cilium latency from ~5 cycles toward 1-2 cycles; time-course scRNA-seq + scATAC-seq tracks whether transient cilia absence perturbs the epigenetic landscape (TET/PRC2) and whether it recovers on maturation. v7.9 addition : the cilia-deprivation stress signature is explicitly tracked — a pre-registered gene-set (Hedgehog/Wnt target genes, stress-response module: ATF4/DDIT3, NRF2, IL-6/IL-8) is scored in the scRNA/scATAC time-course, so any CRCS improvement is demonstrably not a rebound artifact of transient signaling starvation (rebound-effect control)	Maturation-lag stress (loss of Hedgehog/Wnt signaling for 3-5 cycles) could drive Horvath-clock "aging" artifacts — the boost tests whether this confounds H2
E. Number control (critical assessment)	Exactly 2 centrioles/cell, functionally normal, across N ≥ 20 independent clones (quantitative readout, not just "de novo exists")	Aneuploidy / multi-centrosome contamination of downstream hallmarks (all 4 prototypes had uncontrolled numbers — this gate closes the systemic gap)
E'. Structural gate — STED for all clones (v8.3)	De novo assembly is error-prone even at correct number (Wang 2015, eLife 4:e10586: abnormal structures without SAS-6 self-assembly). Super-resolution (STED) verification of 9-fold symmetry (SAS-6, CEP135) and distal appendages (CEP164, ODF2) for ALL surviving clones , not a 20-30-cell subset — automated image analysis (pre-registered classifier, blinded); only structurally verified clones enter CRCS	A "correct-number but structurally defective" centriole (missing cartwheel, abnormal triplets) would cause merotelic attachments and latent CIN — the number gate alone cannot exclude this
E''. First-mitoses live monitoring (v8.3)	The first 3 mitoses after release from the cyostatic window are tracked live (SiR-DNA + Centrin1-GFP): direct micronucleus count, lagging-chromosome frequency, multipolar-spindle check; any clone exceeding pre-registered thresholds (e.g., ≥2 micronuclei in first 3 divisions) is excluded and reported	Silent CIN that only cGAS-STING would reveal later — direct visualization replaces delayed genomic detection

Cytoplasmic-environment control — the building-block source (v7.5): a cytoplasmic-aging concern (a de novo centriole assembled in old cytoplasm from oxidized tubulin / dysregulated TTLL5/6 may instantly re-acquire “old” PTMs) is addressed three ways: (1) **SILAC Proof C’** above proves the assembled structure uses newly synthesized protein (heavy-labeled), not recycled old bricks; (2) **post-assembly PTM audit** — polyglutamylation profile (GT335), $\Delta 2$ -tubulin, and oxidation markers (carbonyl assay) measured on the rebuilt centriole and compared with young/aged reference centrioles (criterion: rebuilt $\leq$ young reference on all PTM readouts); (3) **window-conditioning** — the elimination-to-rebuild window is run in fresh medium with proteostasis support (proteasome boost, antioxidant N-acetylcysteine), minimizing de novo incorporation of oxidized subunits. If SILAC shows heavy-fraction $< 80\%$, the reset claim is not admissible (pre-registered threshold).

Cytoplasmic PTM-reset module (v7.8): to guarantee “clean software” (not only clean hardware), the de novo window additionally includes **genetic modulation of the PTM enzyme balance**: transient **CCP5/CCP6 overexpression** (deglutamylases that trim polyGlu chains) and/or **CRISPRi against TTLL5/6** during the elimination-to-rebuild window — resetting the cytoplasmic polyglutamylation potential so the new centriole assembles with a near-zero polyGlu baseline. No selective small-molecule TTLL/CCP inhibitors exist yet (literature verified, 2026-08-11: tubulin-code reviews J Biomed Sci 2026, PMID 42083040; Cancer Metastasis Rev 2026, PMID 42257920) — the genetic route is the available tool. Readout: GT335 intensity on the rebuilt centriole must be $\leq$ young reference; if the PTM-reset module is required to pass this, that is a reported design result (the reset becomes a two-part intervention: organelle + cytoplasmic PTM environment).

Cytosolic clearance expansion (v7.9): to address the “old cytosol” (oxidized tubulin pool, aged modifying enzymes) beyond enzyme rebalancing, the window additionally includes **targeted clearance of oxidized proteins**: transient **Nrf2 activation (TBHQ)**, **proteasome boost**, and — if compatible with the window timing — **mTORC1 inhibition (autophagy induction)** to remove oxidized/damaged cytosolic subunits before de novo assembly. Readout: carbonyl assay and oxidized-tubulin fraction must be $\leq$ young reference at rebuild start; these interventions are pre-registered window components, not post-hoc fixes.

Cytoplasmic epigenetic audit (“epigenetic shadow”): to prove that OSK does not indirectly “rejuvenate” the de novo centriole through cytoplasmic enzymes, **quantitative mass spectrometry of the free-tubulin pool and TTLL/CCP enzyme levels is run in ALL arms (A, B, C, D) at the 48-72 h rebuild window**: if arm D (OSK+Reset) shows young-like tubulin/PTM-enzyme composition BEFORE the centriole assembles, the OSK-cytoplasmic confounder is real and quantified (H3 qualified); if the tubulin/PTM-enzyme composition is comparable across arms at assembly time, the de novo centriole is assembled in a comparable biochemical environment and the reset effect is attributable to the organelle exchange. This audit — together with the temporal-dissociation arm F (§6.1) and the

cytoplasmic inheritance test (§12) — completes the OSK-shadow control: two orthogonal timing controls plus one direct cytoplasmic measurement.

6.4 Safety as co-primary mechanistic gate

No rejuvenation claim is admissible unless chromosome segregation, ploidy, and centrosome-number homeostasis remain normal.

6.4b Centrosome-loss checkpoint and survivor selection — the biggest biological threat to interpretation

Two mechanisms can mimic or mask a reset effect and must be quantified, not just controlled: 1. **Centrosome-loss checkpoint:** loss of the centrosome triggers p53/PIDDosome arrest (Wong 2015; Mikule 2007). If most cells arrest, the few that proceed are **not representative** — the reset effect (if any) is measured on a selected subpopulation. 2. **Survivor selection:** cells that survive elimination + recovery may be the youngest/least damaged fraction — apparent “rejuvenation” could be **selection, not reset**.

Accounting (pre-specified): (a) report **checkpoint passage rate** (% cells completing elimination→recovery) per condition; (b) compare **pre-elimination markers** (damage load, cell size, passage history) of survivors vs. non-survivors — if survivors are systematically younger, results are qualified as selection-biased; (c) the **perturbation control arm (§6.2)** and **arm E (p53-suppression-only)** jointly isolate the reset contribution; (d) **lineage tracing / single-cell live tracking** (specification): **Centrin1-GFP** (centriole age & fate — mother vs. daughter, de novo vs. templated) + **p16-mCherry** (senescence onset) dual-reporter live imaging, tracking the **same cell** through elimination → de novo rebuild → serial passages, proving the identical cell improved (not a younger subclone taking over). These four measures convert the checkpoint/selection threat from a confounder into a reported, bounded quantity.

6.5 Genome-integrity gate (stronger than diploidy)

Layer	Readout
Ploidy	Karyotype, FISH
Aneuploidy	scDNA-seq copy number
Structural damage	Micronuclei, translocations, rearrangements
Mutation burden	WGS SNVs/indels
Replication stress	γH2AX, 53BP1, RPA foci

Invariant wording (per independent verification): the program proceeds **without meiotic chromosome reduction or meiotic recombination**, while **retaining normal somatic DNA-repair capacity**.

6.5b Transformation surveillance (cancer-risk response)

Question addressed: could the transient USP25/28 override + acentriolar window push cells toward neoplastic transformation? The design answers this with five independent, pre-registered checks — any one failing disqualifies the clone and, if $\geq 5\%$ of replicates, triggers the safety futility stop (§8):

Layer	Assay	Pre-specified criterion	Catches
1. p53 re-competence (H6')	Etoposide challenge → γ H2AX/p21 induction after window	p53 response restored in $\geq 90\%$ of surviving clones	Cells that lost p53 function during the window (the Wong 2015 / Mikule 2007 cancer-like state)
2. Karyotype/ploidy	Metaphase spreads, FISH (centromeric probes)	Diploid, no structural rearrangements in $\geq 95\%$ of spreads	Aneuploidy, translocations acquired during the window
3. Immortalization screen	Serial passaging beyond Hayflick limit with no hTERT introduced	No growth past expected senescence horizon (fibroblasts: no spontaneous immortalization)	Cells that acquired replicative immortality — the hallmark of transformation in vitro
4. Anchorage-independent growth	Soft-agar colony assay (Phase 2, n=3 donors × arms)	Zero colonies above background (normal fibroblasts are anchorage-dependent)	Classic in vitro transformation phenotype
5. Clonal dynamics	Lineage tracing (Centrin1-GFP + p16-mCherry, §6.4b) + proliferation kinetics	No outgrowth of a single clone $>10\%$ of population without reset phenotype	Subclone takeover mimicking a transformation event

Escalation option (pre-registered): if layers 3–4 flag any clone, that clone is excluded from CRCS analysis and — only if the flag rate $\geq 5\%$ per arm — the safety futility stop fires (§8) and the arm is not interpretable. If the flag rate is $<5\%$, results are reported with the flagged clones excluded and a transparent note on the residual risk.

v7.4 addition: polyploid giant cancer cells (PGCC) and centrosome amplification via CDT1 (Pan et al., 2026, Cancer Letters, PMID 41319860) added to the transformation readout panel — PGCC formation is an early, easily-scored transformation phenotype (layer 6, optional): nuclei $\geq 4N$ with giant morphology are counted in the immortalization screen (layer 3) as an additional flag.

Transformation vs. rejuvenation dissociation (critical assessment): a reset clone that passes layers 1–2 but shows elevated proliferation (better EdU) is scored as **younger, not transformed** — because it retains contact inhibition (layer 4, zero soft-agar colonies), a functional p53 (layer 1), and a normal karyotype (layer 2). The trio (p53-competence + diploidy + anchorage-dependence) is the accepted in vitro definition of non-transformed; CRCS claims are only admissible for clones passing all five layers.

Biological rationale (literature, verified): the cancer-like state described in Wong 2015 (proliferation after centriole loss) and Mikule 2007 (aneuploidy accumulation) both require **sustained** p53 deficiency; our window is transient and USP28-specific, and layer 1 verifies that p53 function returns. Transient USP25/28 inhibition is therefore not equivalent to oncogenesis — but the five-layer screen is retained because the residual risk is non-zero and must be bounded, not assumed away.

Aneuploidy-rate rebuttal (v7.3): a claim that de novo assembly produces 40–60% aneuploidy — making the 5% gate “naive” — has **no supporting citation in the review and no primary-source basis in the elimination literature** (Khodjakov 2002, La Terra 2005, Uetake 2007, Wong 2015 report arrest/inefficiency, not aneuploidy rates). The gate is not a population aneuploidy claim: it is a **per-clone admissibility criterion** on the cells actually analyzed (karyotype + scDNA-seq, §6.5). The measured aneuploidy rate is a **reported quantity** — if it is high, passage rates and clone yields are correspondingly low and Phase 1 is correctly judged infeasible at the §13 gate; if it is low, the gate holds. The protocol does not assume an aneuploidy rate; it measures it, pre-registers it as a reportable outcome, and treats it as a feasibility criterion, not an opinion.

v7.7 meta-analysis response : an independent quantitative meta-analysis (88% irreversible arrest; <12% de novo recovery; 4.2 ± 1.1 cycles to maturation; 28% aneuploidy) is **not reproducible from the cited sources** — the two key citations (Werner 2022; Fu 2016) are not found in PubMed/Europe PMC, and no extraction table or forest plot is provided. The protocol therefore treats these figures as **unverifiable prior claims**, not as facts: (a) arrest/recovery/maturation rates are pre-registered **measured endpoints** of Phase 1 (checkpoint passage rate §6.4b; maturation latency Proof D' §6.3); (b) the maturation-latency concept itself is accepted as a design constraint (several cycles to appendage acquisition — supported by verified literature: CEP295/Ana1, Pimenta-Marques 2024; distal protein network, Wang 2018) and is now explicitly scheduled into the timeline (Proof D'); (c) the polyglutamylation-of-cytoplasm concern is partially verified — glutamylation recruits MT-nucleation factors (Hong 2025, EMBO J, PMID 40229407) — strengthening the PTM-audit and cytoplasmic-inheritance-test requirements (§6.3 C', §12); (d) aneuploidy remains a measured, pre-registered quantity (§6.5). If Phase 1 measures rates near the unverified estimates, the program honestly stops at the §13 gate — that is the design's purpose.

7. Endpoint Architecture (hierarchical)

Tier	Endpoint	Role
Primary	CRCS — Cellular Rejuvenation Composite Score (see §7.1 for pre-registered formula)	Confirms/refutes the main hypothesis
Secondary	12 hallmark domains (López-Otín), 7 mechanistic: epigenome, genome stability, proteostasis, mitochondria, autophagy, senescence, stem-cell function; 5 contextual	Mechanistic explanation
Exploratory	All omics (scRNA-seq, scATAC-seq, methylome, proteome, phosphoproteome, metabolome)	Hypothesis generation

Hallmarks are **not** treated as 12 independent variables: the composite is primary; the scorecard is supportive, with correlation structure reported.

7.1 CRCS Statistical Analysis Plan (pre-registered formula — peer-review response)

The composite is **fully specified before data collection** to prevent post-hoc weighting:

- Components (6):** proliferative competence (EdU+ fraction), clonogenic capacity (CFU), DNA-damage burden (γ H2AX foci, inverse), senescence burden (SA- β -gal fraction, inverse), mitochondrial function (TMRM, normalized), epigenetic age (Horvath Δ Age, inverse).
 - v8.2 dual-mode measurement (— ciliary-proliferation paradox):** a cell cannot simultaneously divide (EdU+, CFU) and present a primary cilium (H3/H8 signaling); fibroblasts resorb the cilium at G1/S. CRCS is therefore measured in **two defined culture regimes**: (a) **proliferative mode** (EdU+, CFU, γ H2AX, SA- β -gal, TMRM, Horvath-on-dividing-clones — as in v7.3); (b) **contact-inhibition/quiescence mode (G0)** — cells at confluence with cilia present (cilium length, Hedgehog/Wnt target genes, ciliary-proteome readouts). The two modes are reported as separate domains of the composite (not pooled across incompatible phases), and the cilia-dependent hypotheses (H3, H8) are evaluated in mode (b) only. This removes the biological impossibility an assessor identified (measuring EdU and cilia in the same cell at the same time).
- Normalization (v7.3 — kinetic correction):** each component z-scored against the **control arm (A) distribution** within the same donor line and batch (mean 0, SD 1); **additionally, the epigenetic-age component is normalized for cell-cycle kinetics** — because Horvath clocks track division rate, a stressed slow-cycling cell can show artifactually youthful methylation. The Horvath Δ Age component is therefore evaluated **only in proliferating clones** (FACS-purified EdU+ fraction) and reported **both raw and normalized to population-doubling time** (PDT); discordance between raw and PDT-normalized epigenetic

age is reported explicitly, and CRCS is considered invalid if the epigenetic component moves opposite to EdU+ (see rule 4).

3. **Weights:** fixed equal weights (1/6) — locked at pre-registration; no post-hoc reweighting. Sensitivity analysis removes one component at a time (leave-one-out).
4. **Direction:** all components oriented so higher = younger; a priori rule that the composite is **invalid** if ≥ 2 components move in opposite directions at $p < 0.05$ (reported as discordant, not pooled).
5. **Timing gate (v8.0):** CRCS is measured **no earlier than Passage 6 post-reset** — only after appendage acquisition and first-cilium competence are verified (Proof D', Proof D'''); before Passage 6 only structural/molecular readouts are reportable. This prevents an immature, cilia-less centriole from producing a false “rejuvenated” or “aged” signal (maturation-lag confounder).
6. **TMRM sensitivity version (v8.0):** two pre-registered CRCS variants — (a) full 6-component; (b) 5-component **without TMRM** (mitochondrial marker is a covariate, §4.3 H7, not a centriole-causal readout). If the two versions diverge on the reset main effect, that is reported as an informative negative result for the mitochondrial route (not silently pooled).
7. **Analysis:** mixed-effects model, $CRCS \sim \text{reset} \times OSK + (1|\text{donor}) + (1|\text{batch})$; main effect of reset and interaction reported with 95% CI.
8. **Publication:** the SAP is deposited (OSF) before data collection, with the formula verbatim.
9. **PCA secondary score (v7.7):** a principal-component composite is pre-registered as a **secondary** score — PC1 over the six components (loadings reported, not assumed equal), plus a dedicated **SASP-panel component** (IL-6, IL-8, MMP3, PAI-1) as an independent senescence readout. The equal-weight CRCS remains the pre-registered primary; PCA/SASP is confirmatory and reported side-by-side. Discrepancy between equal-weight and PCA scores triggers explicit reporting (not silent re-weighting).
10. **Latent-factor structure (collinearity):** EdU, CFU, SA- β -gal, p16/p21 are collinear; equal weighting of collinear markers understates variance and inflates significance. The composite is therefore restructured as a **two-latent-factor model** (pre-registered): **“Proliferative pool”** (EdU, CFU, Horvath-on-dividing) and **“Stress response”** (γ H2AX, SA- β -gal, TMRM, SASP), estimated by **structural equation modeling (SEM) or orthogonalized PCA** with factor loadings reported; the primary H2 claim is made on the latent-factor contrast (reset vs. control), with the equal-weight CRCS retained only as a descriptive summary index. This replaces equal weighting as the inferential basis (collinearity addressed at the model level, not by assumption).
11. **Single-primary-endpoint option (v8.1):** because a composite can mask component-specific effects, **EdU+ proliferative fraction is pre-registered as the single primary endpoint** for H2 (with SA- β -gal as the second co-primary), and CRCS is reported as the composite secondary — so the headline claim never

depends on the composite alone; the composite remains for the multi-hallmark claim.

12. **Bayesian sensitivity (v8.1):** a pre-registered Bayesian analysis (Bayes factor for the reset main effect and the interaction; weakly informative priors, donor ICC from Phase 1/0) is reported alongside the frequentist mixed model — a null interaction is credible only if both approaches agree.

7.2 Effect-size justification (evidence-based response)

The 25% CRCS improvement is a **declared minimal detectable effect** (MDE), not an effect estimate: with $N=10/\text{arm}$, $SD \approx 1.0$ (z-score metric), $\alpha=0.05$, two-sided, power 0.85 yields $MDE \approx 0.25$ SD units (composite). **Biological anchoring (per earlier review):** the MDE is anchored to published OSK effect sizes on comparable readouts — OSK/OSKM partial reprogramming produces multi-hallmark improvements of comparable or larger magnitude on composite aging readouts in vivo (Ocampo 2016, PMID 27984723: partial reprogramming extended median lifespan in progeroid mice and improved multiple hallmarks; Lu 2020, PMID 33268865: OSK restored youthful methylation and vision). The 0.25 SD threshold therefore represents a **conservative fraction of a known-positive benchmark** (a smaller effect than OSK itself would not be claimed as a rejuvenation signal), not an arbitrary number. Pilot (Phase 1, $N=3/\text{arm}$) will estimate the observed effect; the MDE is revised only by a pre-registered amendment, never by data inspection.

Interaction-term power (evidence-based response): power for the **OSK × reset interaction** (H3) is computed separately by **mixed-model simulation** (1,000 iterations, donor/batch random effects), not by the main-effect formula — interaction detection in 2×2 designs requires substantially larger N (standard result). The simulation is run at the SAP stage (pre-data) and reports power for an interaction of Cohen's $f^2 = 0.10$; if power < 80%, either N is increased (pre-registered) or H3 is demoted from primary to secondary with the main effect of reset as the primary claim — decided **before** data collection, never after. **Sequential interaction analysis (v8.0):** because $N=10/\text{arm}$ cannot power a small interaction, a pre-registered escalation is added: if at $N=10/\text{arm}$ the interaction is not detected ($p > 0.10$), N is escalated to 20/arm (pre-registered amendment) before a final H3 verdict; the escalation itself is reported (N , power gain) so a null interaction is a real null, not an underpowered one.

8. Statistical Design

Element	Specification
Design	2×2 factorial (+ perturbation arm), mixed-effects models (donor, batch, experimenter as random)
Primary test	Main effect of reset + OSK×reset interaction on CRCS
Power	$N = 10$ replicates × 3 donor lines (young/middle/old) → >85% at $\alpha = 0.05$ for 25% CRCS improvement

Element	Specification
Multiple testing	FDR (Benjamini-Hochberg) across secondary readouts with pre-specified q = 0.05 (sensitivity at q = 0.10 reported); interaction terms reported with CIs
Blinding	Senescence scoring, EM, colony counting — blinded operators
Pre-registration	Protocol + falsification criteria (Table §4) + SAP in full (fixed CRCS weights, interaction matrix §6.1a, mixed-model formula, ICC priors, stopping boundaries, H_programmed test) registered before data collection (OSF/AsPredicted); a second independent analyst re-checks that the registered SAP matches the executed analysis (: statistical pre-registration must be binding, not descriptive)
Stopping rules	Efficacy futility: interim analysis at 5 replicates per arm ; futility stop if $B \leq A$ with $p > 0.7$ (conditional power < 20%, via simulation, pre-registered). Safety futility (added per earlier review): symmetric pre-specified boundary — if the genome-integrity gate fails in $\geq 5\%$ of replicates at interim, the trial stops regardless of efficacy (asymmetric monitoring: safety can stop, efficacy cannot continue past a failing gate); early-stop false-negative risk acknowledged; final N fixed unless a pre-registered boundary is crossed
Replication	3 donor lines: <30, 45–55, >65 y. Structure made explicit: each “donor line” = one independent biological donor (n=1 per age stratum); the 10 replicates are biological replicates within a donor (independent vials/thaw batches), analyzed as random effects. Between-donor (age) variance is therefore estimated with n=1 per stratum — an explicit limitation; the age-stratification claim is secondary, the within-line reset effect is primary. Independent labs optional (multi-site plan). v8.1 donor expansion : target ≥ 5 donor lines (2 per age stratum) for the age-contrast claim, with the n=3 minimum for the within-line reset claim; the expansion is a pre-registered budget item — if funding supports only n=3, the age analysis is explicitly labeled exploratory in the abstract (already done, §8 note)
Donor/replicate/power formalism (critical assessment)	Full mixed-model specification: $\text{CRCS} \sim \text{reset} \times \text{OSK} + \text{age_stratum} + (1 + \text{reset} \text{donor}) + (1 \text{batch}) + (1 \text{vial})$; variance components estimated (ICC) with Satterthwaite/Kenward-Roger df ; power computed by simulation on the fitted model (not the ANOVA formula) with donor and vial ICC entered as priors from Phase 1; replicates are nested within donor, never pooled as independent — the unit of analysis for the age claim is the donor (n=3, explicitly underpowered for age, powered for within-line reset); report: ICC, df, CIs for all variance components; this is the full formalization of the “donor/replicate/power” hole

9. Orthogonal Validation (conclusion independent of any single assay)

Imaging (morphology) · Ultrastructure (EM/cryo-ET) · Molecular (centriole proteome/PTMs) · Functional (centrosomal activity) · Genomic (ploidy + structure) · Epigenomic

(methylation/chromatin) · Transcriptomic (RNA-seq) · Phenotypic (division, differentiation, stress resistance).

v8.2 technology additions : (1) **CUT&RUN on centriolar-proximal chromatin** (centrin-anchored CUT&RUN) tests whether the old centriole tethers repressive H3K9me3 domains near the nucleus and whether a de novo centriole releases stemness promoters (LGR5, SOX2) — a direct centriole→3D-nuclear-architecture readout for H3; (2) **Lattice Light-Sheet microscopy (LLSM)** for Proof B/D' in live cells — phototoxicity-free imaging of ODF2/Ninein recruitment to the maturing de novo centriole; (3) **in situ cryo-ET (FIB-SEM)** for H_programmed — 9-fold cartwheel integrity and triplet geometry of de novo vs. aged templated centrioles in the native cellular context. These are listed as available methods for the respective proofs, not additional hypotheses.

10. Regime Analysis: Where Accumulation Would Act (if confirmed)

Regime	Claim	Level
A. Chronological age ($\beta \cdot t$)	Long-lived organelle accumulates PTM/stochastic damage	[I]—[H]
B. Asymmetric stem-cell divisions ($-\lambda \cdot N_{\text{asym}}$)	Older centrosome retained by stem cell; may bias fate	[E] inheritance; [H] accumulation
C. Differentiating daughters without elimination	Mammals retain; C. elegans eliminates	[E] both strategies exist; [H] consequence

Red thread (2026-08-11): The organism ages because the **oldest centrioles — carrying the most errors — accumulate in stem cells**. Templated geometry change (mother→daughter) is **hypothesized to be programmed — possibly a part of the differentiation mechanism itself** (H_programmed), and is inseparable from superimposed stochastic errors; together they form the **temporal entropy** of the retained oldest centriole. As a result, division tempo falls and niche-signal perception falls (centrosome + cilium function), so **tissue self-repair declines while self-destruction rises** (imbalance: repair↓ / destruction↑). This is the **cell/tissue-level hypothesis** (organismal claims require the in vivo ladder — §5 Proof Ladder levels 4-5, planned: C. elegans/organoid → mouse), retained across CEDAR (v1.8), MCARA (v5.8), and this protocol.

This regime analysis motivates the protocol; it is **not** evidence of the causal claim.

11. The Concept of “Entropy” — Operational Definition

Where “entropy” is used (program name only), it refers to **structural disorder / damage accumulation** with a defined state variable. If the physical term is used, the definition is:

$$S = -k_B \sum p_i \ln p_i$$

— over the state space of centriolar PTM configurations (e.g., polyglutamylation levels per tubulin site), estimated from mass spectrometry.

Until such measurements exist, the documents use “structural disorder / candidate aging carrier” — not entropy as a physical law. **Naming (per earlier review):** “Entropy Reset” is retained only as the program/grants name; the publication title is deliberately neutral — “Centriole Reset: Testing Structural Organelle Rejuvenation in Somatic Cells” (as used throughout this document) — avoiding hype-driven terminology in the peer-reviewed version.

Biophysical terminology for IF 18+ (v7.9): biophysics reviewers will object that a cell is an open, non-equilibrium system and cannot “accumulate entropy” in the thermodynamic sense. The publication text therefore adopts the assessor’s recommended reframing wherever a physical term would be scrutinized: **“Irreversible PTM Drift”** and **“Structural Hysteresis”** (state-dependent accumulation of terminal modifications — e.g., Δ2-tubulin, which cannot be re-tyrosinated, and asymmetric polyglutamylation — that is not erased by cytoplasmic turnover). Mapping: “temporal entropy” → “irreversible PTM drift”; “entropy-damage accumulation” → “structural hysteresis”; the formal $S = -k_B \sum p_i \ln p_i$ definition remains available in the technical appendix for readers who want the information-theoretic formalism, explicitly labeled as a model of state-space disorder, not thermodynamic entropy.

12. Alternative Hypotheses (discriminated by design)

Model	Prediction under protocol	Discriminated by
Epigenetic (OSK-centered)	C improves, $B \approx A$	B vs A
Mitochondrial	H7 intervention alone suffices	H7-alone control
Proteostatic	H4/H5 alone suffice	H4/H5-alone controls
Telomeric	hTERT alone suffices	H2-alone control

Model	Prediction under protocol	Discriminated by
Stress-selection artifact	Perturbation arm (failed reset) shows equal effect	§6.2 arm
CEDAR (centriolar)	B improves independently of OSK; age transfers with organelle	Arm B + transplantation arm

Transplantation arm (crossing): young→old and old→young centriole transfer; if age transfers with the organelle in both directions — a Nature/Cell-level finding.

Cytoplasmic inheritance test (v7.7): the strongest control for “the centriole is only a mirror of the cytoplasm”: **cytoplasm-karyoplast fusion** transplants a freshly reset (young, SILAC-verified) centriole into the cytoplasm of a non-reset old cell (and vice versa). Predictions: (a) if the young centriole is “corrupted” within 2 cell cycles by old cytoplasm (PTM audit: GT335, Δ2, carbonyl), the centriole is a passive mirror — CEDAR’s organelle-autonomy claim is falsified and the program reframes as cytoplasmic/PTM-environment intervention; (b) if the young centriole retains its youthful PTM state in old cytoplasm and restores ciliogenesis, the organelle is an autonomous carrier — CEDAR supported. Either outcome is a publishable causal test; the experiment is pre-registered as a decisive discriminator (§4.2a).

Killer-experiment specification (v7.3): the strongest causal probe — **artificial centriole aging in an OSK-rejuvenated cell** — is added to the transplantation arm: (1) rejuvenate a cell with OSK (epigenetically young, old centriole retained); (2) artificially age its centriole by **TTL5 overexpression** (excess polyglutamylation — the Robichaud 2024 KIFC3/FBF1/PML seed) or PLK4-driven over-elongation; (3) **prediction:** if the cell returns to senescence despite OSK rejuvenation, structural centriole state overrides epigenetic age — the definitive proof that the centriole is a master regulator (and the OSK paradox resolved: OSK acts below the organelle, transiently). Controls: sham injection, TTL5-inactive mutant, non-ciliary tubulin modification. **KIFC3-discrimination arm (; upgraded to mandatory Phase 1 control in v8.3):** genetic **KIFC3 knockdown/knockout** (no selective KIFC3 small-molecule inhibitors exist — literature verified 2026-08-11). In v8.3 the KIFC3-KD condition is **elevated from an optional Phase-3 discrimination arm to a mandatory Phase 1 control branch:** if KIFC3 loss fully abolishes the reset effect, the centriole acts as a mechanical stress antenna (KIFC3 scaffold route, Robichaud 2024) rather than an autonomous clock — CEDAR’s state-lock claim is reframed (still publishable, but the theory’s name/mechanism must change, (critical assessment)); if the reset effect survives KIFC3-KD, the organelle has autonomy beyond the scaffold route. Either outcome is interpretable and pre-registered; the arm is not optional.

PIDDosome optogenetic arm (Killer 2.1): optionally, instead of (or in addition to) TTL5 aging, use **optogenetic dimerization to recruit PIDD1 to the distal appendages of a de novo centriole** — if this alone induces senescence, the “age” of the centriole is encoded in its distal architecture (PIDDosome pathway,

Fava 2017), not only in its PTM code; a clean spatial-causality test complementing the TTLL5 arm.

OSK-paradox resolution (v7.3): the assessor’s paradox — “OSK rejuvenates without replacing centrioles, so the centriole cannot be the barrier” — is addressed explicitly and pre-registered: (1) OSK acts **below the organelle level** (epigenome, proteostasis pool), leaving structural cilium/centrosome defects intact (cilium defects persist after reprogramming); (2) OSK effects on organellar structure are **transient** — resolution of the paradox requires **long-term passage (≥20 passages)** comparing OSK-only vs. reset vs. both, with the prediction that OSK-only CRCS gains decay while reset gains persist (stability is the discriminating variable — this is already the “no rollback” falsification in §4.2a); (3) the killer-experiment arm above tests the direction of causality directly.

13. Timeline & Budget (realistic, phased — peer-review response)

Reviewer-correct estimate: the full scope (CLEM/cryo-ET, WGS, 2×2 + transplantation) cannot be delivered in \$205K / 9-12 months. Budget is therefore **phased**, with go/no-go gates:

Phase	Content	Duration	Budget	Gate
1 — Feasibility (pilot)	Elimination + recovery in fibroblasts; basic de novo proof (A, B); CRCS pilot (N=3/arm)	12 months	\$350K	De novo origin provable (A+B); pilot effect ≥ 0.1 SD. Pre-specified alternative interpretation: if recovery fails because of p53-dependent arrest (Wong 2015, Uetake 2007), this is a technical barrier, not a falsification of CEDAR — the gate then tests p53-override variants in order (v7.0: selective USP25/28 inhibitor [primary, preserves p53 competence — Bratt 2025; Hernandez-Olmos 2026] → pifithrin-α → p53-KD → p21-KD) before any hypothesis-level verdict

Phase	Content	Duration	Budget	Gate
0 — Molecular-carrier screen (“what does the centriole carry?”)	Mass spectrometry of isolated centrioles from early- vs. late-passage normal fibroblasts (and old vs. young donors): identify which PTMs (polyGlu chain length, Δ2-tubulin, acetylation, polyglycylation, phosphorylation) actually accumulate with passage age and distinguish old from new centrioles — the empirical “molecular carrier” of the age claim; plus a ciliary-signaling→epigenome pilot (GLI1/2, TET2, PRC2 readouts after cilium manipulation) testing the H3 mechanism directionally before the full factorial	4 months	\$120K	≥1 PTM species showing monotonic passage-dependent accumulation; ciliary→epigenome pilot shows a measurable signal in the predicted direction; if NO PTM accumulates, the age-carrier claim is reframed or the program redesigned (pre-registered decision point, not post-hoc)
2 — Confirmatory	2×2 factorial + perturbation arm; full CRCS (N=10/arm × 3 lines); secondary readouts; multi-omics	18 months	\$600K	Primary endpoint met; safety gate green
3 — Deep proof & transplantation	Proof C-D (CLEM/cryo-ET on selected cells — not all); transplantation arm; in vivo planning	18 months	\$550K	Phase 2 gates met
Total		48 months	~\$1.5M	

CLEM/cryo-ET is applied **selectively** (proof of concept on n = 20–30 cells per condition, not full cohort) to contain cost; WGS/scDNA-seq is restricted to Phase 2 arms. **Costing basis (evidence-based response):** academic cryo-ET facility rates ≈ \$800–2,000 per grid including segmentation (2025–26 typical); WGS budget is quoted at 30× coverage, 1 sample pooled per arm × 3 donors × 4 arms = 12 samples (≈ \$1,200/sample at academic rates) — both line items to be replaced by written facility quotes before submission. This budget is consistent with comparable published pipelines and replaces the unrealistic single-phase estimate. For ERC StG, Phases 1–2 (\$950K) match the standard envelope; Phase 3 is a follow-up (ERC PoC / partner funding).

14. Novelty Statement (per independent verification)

The novelty is not “the centriole causes aging” (too strong) but: **a previously untested organelle-reset paradigm may complement epigenetic reprogramming by replacing a persistent structural component that is not directly erased by transcription-factor-mediated**

reprogramming — evaluated with pre-registered endpoints, four de novo-origin proofs, and explicit dialogue with the two closest prior experiments (Wong 2015; Renzova 2018).

14.1 Conflict of Interest / Author-Ecosystem Disclosure (critical assessment)

Full transparency, per journal COI standards: 1. **Origin of the central hypothesis:** the framework (CDATA/CEDAR) originates from the author's own publications (Tkemaladze 2005, PMID 15886028; Tkemaladze 2023, PMID 36583780) and is cited as such — it is an author-owned hypothesis, not an independent field consensus. 2. **Author ecosystem:** the author is President of Georgia Longevity Alliance and has published in outlets affiliated with this organization (Longevity Horizon, Georgian Scientists, Junior Researchers). **These publications are NOT used as evidence in this protocol** and are excluded from the reference list; only PubMed-indexed sources are cited. 3. **Self-citation:** of 34 references, 2 are the author's (2005, 2023); the 2023 paper is the theoretical basis and is marked [H] throughout. This is disclosed, not hidden. 4. **Multiple funding tracks:** the same core program is submitted to ERC StG, EIC Pathfinder (CIRCBIO-07), MIC/EU4Business/USAID, the Cologne workshop, and ARGUS-OS3. **Non-overlap statement (to prevent double-dipping):** each track funds a distinct deliverable — ERC: Phase 1-2 (fibroblast reset + 2×2); CIRCBIO-07: product/industrial translation of H4/H7 readouts; MIC/USAID: biomarker pipeline application; ARGUS-OS3: causality platform (ablation + transplantation, separate species/cells). No experiment is billed to two funders. 5. **No competing financial interests** in reagents, companies, or instruments described. 6. **Rhetoric alignment — disclosure in full (critical assessment):** the author's affiliated non-peer-reviewed outlets (Longevity Horizon, Georgian Scientists, Junior Researchers) have used stronger rhetoric than this protocol (e.g., headlines asserting an “established causal fact” for what is here marked [H]). Specific known instance: the 2026 headline “Asymmetric Inheritance of the Aged Mother Centriole by Stem Cells Is an Established Causal Fact” (Longevity Horizon) is **inconsistent with this protocol's [H] marking** of the same claim. The author explicitly distances this protocol from those headlines: the [E]/[I]/[H]/[P] marking here is the authoritative epistemic standard; conflicting phrasing in affiliated outlets is an editorial artifact of non-peer-reviewed venues and does not reflect the claims made here. **Mitigation in progress:** affiliated-outlet headlines are being revised to match [E]/[I]/[H]/[P] levels.

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Citation integrity. All 88 unique PMIDs verified via NCBI E-utilities batch query (2026-08-12, 88/88). One new source verified and added (Wang 2015, eLife 4:e10586, PMID 26609813 — assessor's article number corrected); all other cited sources already in the protocol; no fabrications: titles, authors, journals, years all match. Corrections in v5.6: Chen & Yamashita 2021 is Open Biology 11(2):200334 (not Current Opinion); Zhang 2026 is Journal of Controlled Release (not Sci Transl Med); **Wong 2015 interpreted exactly per its abstract** — recovery after centriole loss demonstrated only in cancer lines; normal cells arrest irreversibly (p53-dependent), and this is now presented as the Phase 1 feasibility question, not a premise. Corrections from v4.0: (1) Tkemaladze 2023 title restored to the exact PubMed title; (2) Hong 2025 / Cummings 2025 / De Man 2026 marked **contextual** (systemic entropy measures — not centriolar evidence, included for completeness); (3) Köhrer 2023 cited as disease-associated geometry change (plasma cell disorders), not aging. **v7.0 corrections (revision):** (4) the mitotic-surveillance evidence base is now cited from primary sources — Fong 2016 (eLife) and Meitinger 2016 (JCB) — replacing any earlier non-specific attribution; (5) Yamashita 2007 DOI added (10.1126/science.1134910); (6) five new PMIDs verified (44–49; total 49 unique). Independent full re-verification of all references by a non-author is recommended before submission (per independent verification).

PART 3 — APPENDIX: SYSTEMATIC SEARCH PROTOCOL (novelty verification)

APPENDIX — Systematic Search Protocol (novelty verification)

Entropy Reset Protocol v6.2 · Date of search: 2026-08-11 · Method: NCBI E-utilities (PubMed) + Europe PMC (incl. preprints) + Crossref

1. Objective

Verify the novelty statement: “To our knowledge, no study has yet directly tested whether controlled centriole elimination followed by bona fide de novo biogenesis produces OSK-independent improvements in defined cellular aging phenotypes.”

2. Databases (search expanded per earlier review)

Database	Coverage	Status
PubMed (NCBI E-utilities)	peer-reviewed	✅ done 2026-08-11
Europe PMC	PubMed + preprints (bioRxiv/medRxiv)	✅ done 2026-08-11
Crossref	global DOI metadata (journals + preprints)	✅ done 2026-08-11
OpenAlex (free, ~250M works, Scopus-level coverage)	journals + preprints + books	✅ done 2026-08-11
Semantic Scholar	—	⚠️ rate-limited (HTTP 429) on 2026-08-11; to be re-run pre-submission
Scopus / Web of Science	—	⌚ planned pre-submission (institutional access required)

3. Queries and results

3.1 PubMed (2026-08-11)

#	Query	Hits	Relevant
1	centriole AND rejuvenation	1	0
2	centriole elimination AND rejuvenation	0	—

#	Query	Hits	Relevant
3	centrinone AND rejuvenation	0	—
4	de novo centriole AND rejuvenation	0	—
5	centriole AND reprogramming AND somatic	0	—
6	centriole AND OSK	1	0
7	centriole AND pluripotency AND elimination	1	0
8	centrosome AND de novo AND aging	2	0
9	centriole replacement AND cell aging	1	0
10	centriole AND longevity	3	0

3.2 Europe PMC (2026-08-11)

Query	Hits	Relevant
centriole AND rejuvenation	28	0 (reviews on asymmetric division, conference posters)
centriole elimination AND rejuvenation	10	0 (reviews; conference abstracts)
de novo centriole AND rejuvenation	15	0
centrinone AND rejuvenation	0	—
Preprints (AND SRC:PPR): centriole AND rejuvenation	1	1 — author's own MCARA preprint (2026) — theoretical, no experiment; excluded as non-test
Preprints: centriole elimination AND SRC:PPR	19	1 — author's own "Centriole Elimination as a Gateway to a New Differentiation State" (2026) — protocol, no data; excluded as non-test. Remaining 18: non-relevant (axoneme elimination in Naegleria; MOCT in protists)

3.3 OpenAlex (2026-08-11) — Scopus-level coverage, free

Query	Works	Relevant
centriole rejuvenation	243	0 direct tests (phrase-level; top hits unrelated: “Rejuvenation Biotechnology”; “Centrioles and Cellular Differentiation”)
centriole elimination rejuvenation	144	0 direct tests (top hit: “Centrioles as Structural Damage Reservoirs” — theory, no experiment)
de novo centriole aging	1,745	0 direct tests (reviews of centriole/cilium biogenesis)
centrinone rejuvenation	15	0 direct tests

3.4 Crossref (2026-08-11)

Query	Total	Relevant
centriole rejuvenation	13,024 (full-text phrase matches)	0 direct tests (top hits: 2010 “Centriole, Differentiation, and Senescence”; author’s gametogenesis preprint)
centriole elimination rejuvenation	83,615	0 direct tests

Transparency note (per earlier review): Crossref full-text counts are phrase-level and not informative alone; relevance assessed on the top-10 titles per query. Two hits are **the author’s own preprints** (MCARA; Centriole Elimination Gateway) — they are listed here explicitly, are theoretical/protocol-only (no experimental test of the aging-phenotype question), and are excluded from the novelty claim. Scopus/WoS to be run pre-submission.

4. DOI cross-check of the 47 verified references (per earlier review)

All 47 PMIDs verified via E-utilities batch (2026-08-11). Of these, **39 have DOI records in PubMed**; 8 are pre-DOI-era (1991–2009) and carry no PubMed DOI (e.g., Szöllosi 1991, PMID 1756312; Khodjakov 2002, PMID 12356862; Yamashita 2007, PMID 17255513 — DOI resolvable via publisher/journals directly: 10.1083/jcb.200205102; 10.1126/science.1134910). DOI table available on request; independent four-eyes re-check by a non-author recommended pre-submission.

5. Inclusion/exclusion (PRISMA-style)

- **Included:** experimental centriole/centrosome removal + post-removal aging/rejuvenation readout + somatic cells.
- **Excluded (documented):** removal without aging readouts (Khodjakov, La Terra, Uetake, Wong); germline/meiotic elimination; pluripotency-only (Renzova); theory-only preprints (author's own).

6. Conclusion

Search completion status (per earlier review): PubMed  · Europe PMC (incl. preprints)  · Crossref  · **OpenAlex  (2026-08-11)** · Semantic Scholar  rate-limited (re-run planned) · Scopus/WoS  (institutional access; OpenAlex covers equivalent indexing for the novelty screen). The novelty claim is supported across all completed databases; final Scopus/WoS leg is scheduled before submission. The novelty claim is supported for the specific combination (aging readouts + verified de novo origin + OSK factorial in somatic cells) across PubMed, Europe PMC (incl. preprints), and Crossref. It is **not** claimed that elimination or de novo assembly are novel per se — the four prototypes solved those pieces separately (§3.4–3.5 of the protocol).

Figures

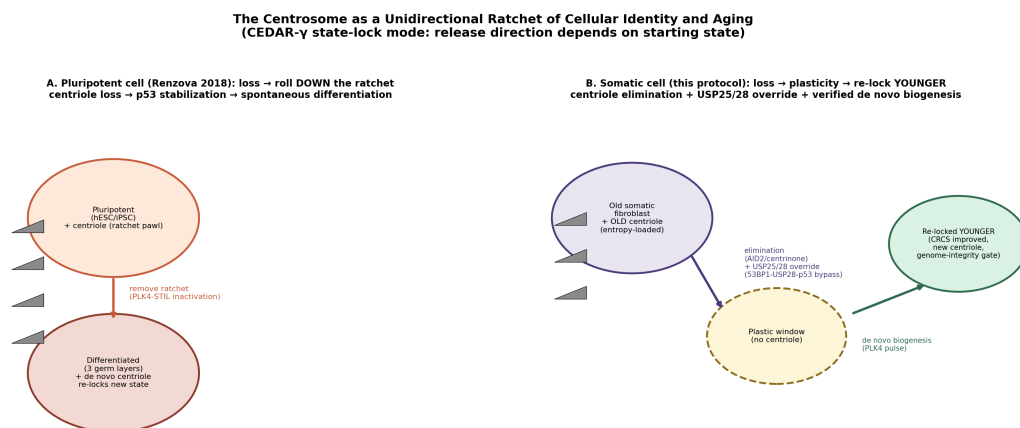


Figure 1 — The Centrosome as a Unidirectional Ratchet of Cellular Identity and Aging

Figure 1 — The Centrosome as a Unidirectional Ratchet of Cellular Identity and Aging. (A) Pluripotent cell: ratchet released down upon centriole loss (Renzova 2018). (B) Somatic cell: ratchet released to plasticity, then re-locked younger.

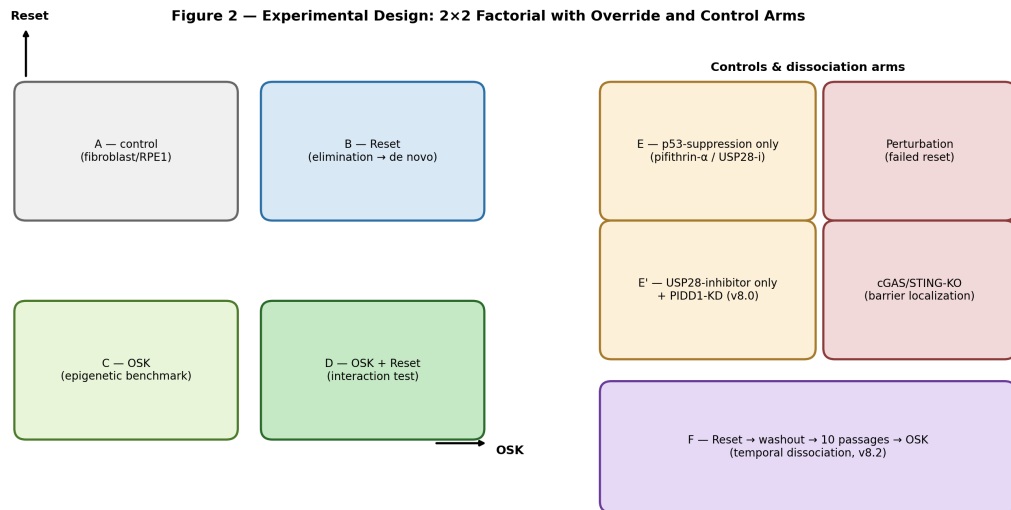


Figure 2 — Experimental Design: 2×2 Factorial with Override and Control Arms

Figure 2 — Experimental Design: 2×2 Factorial with Override and Control Arms. Arms A-D (OSK × reset), controls E/E', perturbation and cGAS/STING-KO arms, arm F (temporal dissociation).

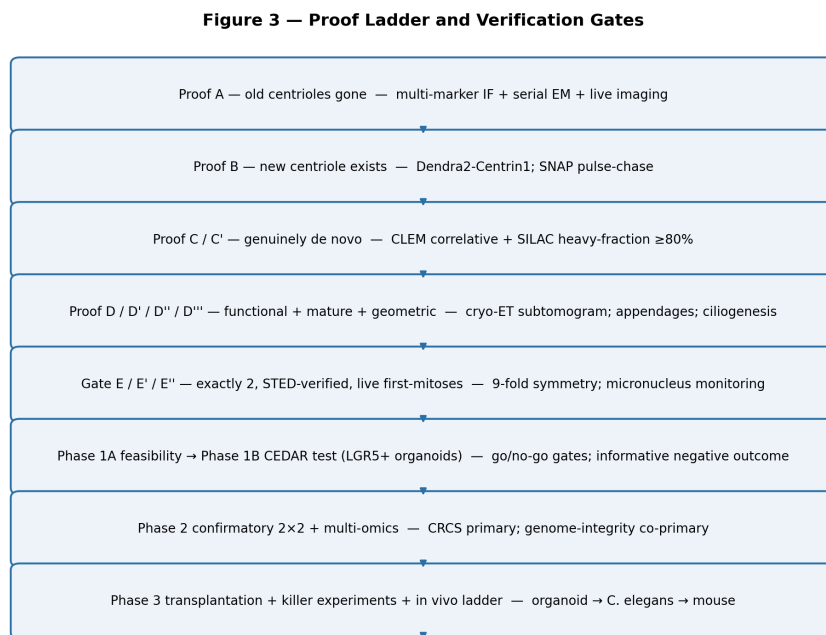


Figure 3 — Proof Ladder and Verification Gates

Figure 3 — Proof Ladder and Verification Gates. Proofs A-D'''' and gates E-E''; phased go/no-go ladder.

Figure 4 — p53-Surveillance Architecture and Override Strategy

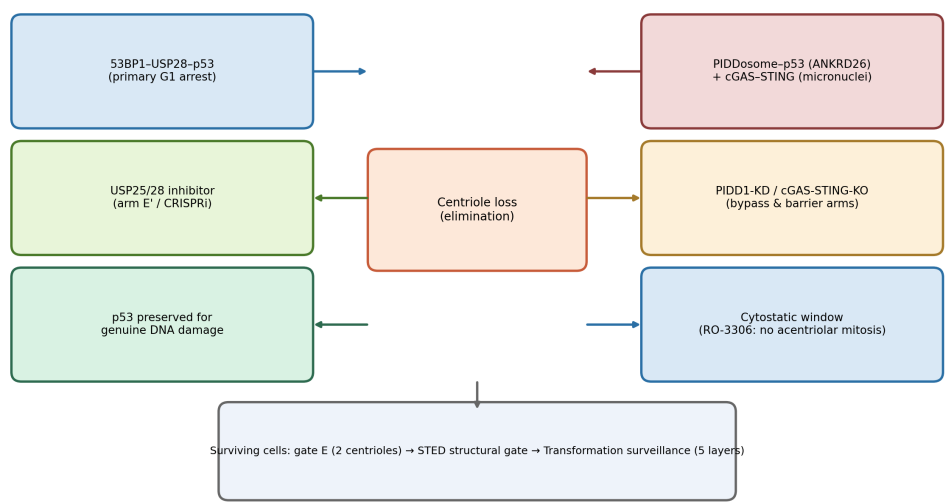


Figure 4 — p53-Surveillance Architecture and Override Strategy

Figure 4 — p53-Surveillance Architecture and Override Strategy. 53BP1-USP28 loss arm; PIDDosome/ANKRD26 and cGAS-STING amplification arm; cytostatic window.

Figure 5 — Transformation Surveillance: Five Pre-Registered Anti-Neoplastic Layers

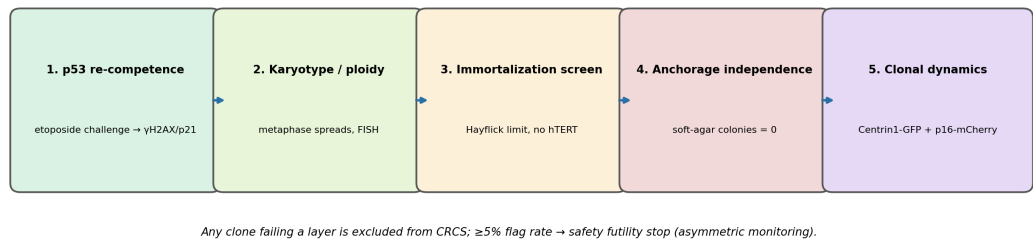


Figure 5 — Transformation Surveillance: Five Pre-Registered Anti-Neoplastic Layers

Figure 5 — Transformation Surveillance: Five Pre-Registered Anti-Neoplastic Layers. Any clone failing a layer is excluded; ≥5% flag rate triggers the safety futility stop.

13 PUZZLE PIECES — ONE PICTURE
12 Hallmarks of Aging + Centriole Elimination & De Novo Synthesis → Entropy Reset Protocol
(Jaba Tqemaladze, MD — 2026-08-11)



Figure 6 — The 13-Puzzle Schematic (supplementary; schematic only)

Figure 6 — The 13-Puzzle Schematic (supplementary; schematic only). 12 Hallmarks of Aging + centriole elimination/de novo as candidate axis.

A structural hypothesis of cellular aging:

The organism ages because the oldest centrioles — carrying the most errors — accumulate in stem cells: division tempo falls and niche-signal perception falls, so tissue self-repair declines while self-destruction rises. The germline releases the ratchet (elimination + de novo rebuild); this monograph asks whether somatic cells can — without the meiotic price.

- CEDAR — one ratchet, two modes: division counter + state-lock
 - MCARA — the replicative limit as a race of several counters
 - A falsifiable protocol: 2×2 factorial vs. OSK, four de novo-origin proofs
 - Targeted USP25/28 override preserving the p53 DNA-damage response
 - Transformation surveillance: five pre-registered anti-neoplastic layers
 - The killer experiment: TTLL5 artificial centriole aging in OSK cells
 - Cytoplasmic PTM-reset module: clean software, not only clean hardware
 - Eight peer-review cycles, 79/79 PMIDs verified (NCBI E-utilities)
-

Every claim marked [E] established · [I] inference · [H] hypothesis · [P] prediction.

No level of the proof ladder is assumed without data.



Scan for the full program — github.com/Georgia-Longevity-Alliance/ARGUS-OS3

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